What Prices Reveal About Hidden Trader Populations, and What Identifying Them Costs

Pratik Niroula* Prashant Niroula† Sujan Khatiwada‡

This version: 30 October 2026

Abstract

Can market prices tell apart groups of traders who react to past prices at different speeds? In a linear market in which groups follow prices with exponentially fading memory and their combined orders move the price, prices reveal only one combined summary of the groups. Groups with different memories can be told apart, but the cost rises steeply as their memories approach: for K unknown groups whose memories are $\Delta\eta$ apart, the number of controlled experiments grows as $(\Delta\eta)^{-(4K-2)}$, with matching bounds on suitably rich classes, and watching without trading needs at least as many observations. For a known pair, the trading capital needed rises sixteen-fold each time their difference is halved. Records that report each group's flow separately keep the problem well conditioned, and a stylised supervisory decision shows when that matters. In pre-specified tests, published trader-class positions in eight CFTC futures markets, and Binance class data in 29 of 29 new markets, pin down a position persistence that prices leave open, and the predicted rates hold with real Bitcoin price noise where it can be used directly. We do not show that this persistence is learning memory, and several pre-specified tests failed, including a test of what prices do reveal, a test of CFTC class flows, and a test that tried to separate learning memory from position persistence.

Keywords: algorithmic trading, identification, trader heterogeneity, market microstructure, system identification, minimax sample complexity, value of disclosure, audit trails.
JEL: C14, C32, C58, D83, G14, G18.

*Department of Computer Information Science, Minnesota State University, Mankato, MN 56001, USA. pratik.niroula@mnsu.edu.

†Faculty of Management, Tribhuvan University, Kirtipur, Nepal. prashantniroula20@gmail.com.

‡Department of Electrical, Computer and Robotics Engineering, Minnesota State University, Mankato, MN 56001, USA. sujan.khatiwada@mnsu.edu.

Code and data. All code, pre-specified test specifications and result files are public at https://github.com/Pratikn03/Fianince.

Reproducibility. Every number in the paper is produced by a script or notebook cell that writes a result file from public data or from a closed form; Supplement S23 maps each number to its source. "Pre-specified" means that the authors committed the specification to the public repository before opening the test data; this is the authors' declaration, not independent preregistration. The proofs have not yet been independently reviewed. Lean 4 with Mathlib machine-checks, using only Lean's standard axioms, the identification theorem, the matching lower and upper bounds for K cohorts and five further results; Supplement S20 states exactly what is and is not machine-checked.

1 Introduction

Markets are increasingly run by algorithms. Trend-following funds, market-making programs, execution algorithms and, more recently, AI trading agents all read recent prices, decide what to trade, and by trading move the prices they read. When something goes wrong, a sharp sell-off, a sudden loss of liquidity, a crowded trade unwinding, the first questions supervisors and risk managers ask are who was trading and how. Which group of traders drove the move? How quickly do they react, and how long do they remember what prices did? Could the same group do it again?

This paper asks how much of that can be learned from what everyone can see, prices and anonymous order flow, and, when public data are not enough, what it costs to find out in other ways.

The question matters now. Trading is passing to learning algorithms and AI agents (Colliard et al., 2022; Dou et al., 2025), and regulators have begun to require firms to say which algorithm sent each order. Under MiFID II, EU trading venues must keep order records that identify the person or algorithm responsible for each investment and execution decision (European Commission, 2017; ESMA, 2016). The US Consolidated Audit Trail (CAT) is designed to record each order in US-listed stocks and options, and the account behind it, across exchanges and broker-dealers; its revised 2026 budget, published in March 2026, is about $147 million (SEC, 2012; CAT LLC, 2026). Such systems serve many purposes, among them enforcement, rebuilding market events after the fact, and surveillance. One question they raise, and the one we study, is how much of the information about who drives price moves could instead be learned from public data, and at what cost.

The short answer. Prices tell an observer less than one might hope, and sharpening what they do tell is expensive.

- Prices reveal one combined summary of all the trader groups, not the groups themselves. Two groups that react to prices with the same memory look exactly like one larger group, however much price data is collected (Theorem 3.1).
- Groups with different memories can be told apart in principle, but the data needed grow very fast as their memories get closer. For two unknown groups, halving the difference between their memories multiplies the data needed by 64; for three groups, by 1,024 (Theorems 5.3 and 5.4).
- Records that report each class's flow separately remove this problem. Each group then becomes an ordinary estimation problem that stays well conditioned however similar the groups are (Proposition 6.1).

We keep three statements apart throughout: what cannot be learned at all, what can be learned but only at high cost, and where estimates become unstable.

Why prices hide the groups. The reason is the feedback loop. Algorithms trade on recent prices, their orders move prices, and those prices feed back into the algorithms. We model each group as tracking the price with a fading memory: recent prices count most, older prices count less, and one number, the group's memory rate, sets how fast the past fades. A desk behaves exactly this way if it uses a Kalman filter or a regularized online learning rule (Lemma 2.1). Because every group reaches the price only through the sum of all their trades, the price records the total and loses the parts. Think of several people pushing the same door: from the door's movement one can learn the total push and how it builds and fades, but two people who push in the same rhythm look like one stronger person.

Three ways to learn more, and what each costs. An observer that wants to know more, whether a supervisor, an exchange's surveillance team or a large trading desk, has three options (Section 2.1).

(a) *Wait and watch prices.* This costs time. In a stationary Gaussian market observed without error, a passive observer needs at least as many observations, in order of magnitude, as an active one needs experiments (Theorem 5.6).

(b) *Trade against the groups* and watch how the price responds, much as one taps a wall to find a hidden beam. Every test trade moves the price against the trader, so learning this way costs money. To tell one group from two groups whose memory rates differ by 2δ, the quantity traded grows like $1/\delta^2$ and the capital at risk like $1/\delta^4$: halve the difference and the capital needed rises sixteen-fold (Theorems 4.1 and 4.3). If memory is a general mixture of fading memories rather than a few exponentials (with the slowest bounded away from a random walk), accuracy improves only with the logarithm of the data, so halving the error squares the data needed (Theorem S17.2).

(c) *Acquire labelled records*, such as audit-trail data or class-level position reports. This costs whatever the reporting system costs, but it removes the explosion in data: labels save a factor of order $(\Delta\eta)^{-(4K-2)}$ at fixed accuracy, where K is the number of groups and $\Delta\eta$ the spacing of their memory rates (Theorem 6.2).

Proposition 6.3 says when each option is the cheapest.

How large are the costs? A hypothetical illustration. To give a sense of scale we price the trading option in a hypothetical benchmark. The model parameters are illustra-

tive; only the size of the measurement noise is estimated, from Bitcoin data. An outside participant that re-learns two unknown algorithms each week with test trades would lose at least $15 million to $356 million a year per market; with three unknown algorithms it would lose more than $40 million a year per market even when their memories are far apart (Supplement S8). These are lower bounds for one stated way of learning, not estimates of what any regulator spends: regulators do not trade to learn. They are best read as the price of the information that labelled records deliver without any trading.

Why this matters for finance. The results point in a constructive direction for several groups of readers.

- *Supervisors and policymakers* get a way to decide what data to require. If the goal is only to watch overall price risk, public prices are enough, because the combined price response is learned at the ordinary rate (Remark 5.2). Labelled records earn their cost when supervision must be targeted, for example when one algorithm must be paused while the others keep supplying liquidity, and when the reporting system covers enough markets to share its fixed cost (Proposition 6.3). In our stylised cost model this illustrates a possible economy of scale for market-wide reporting; it does not explain the legal design of regimes such as CAT. For the narrow task of attributing behaviour to classes, a persistent pseudonymous label is enough, which bears on how much personal data such a task needs but not on what enforcement needs (Section 7). Proposition 6.4 gives a stylised decision in which this matters: the size of a market-wide measure depends only on the combined response, while deciding whether halting one group is enough needs that group's own intensity and inherits the full cost of separating the groups.
- *Exchanges and trading venues*, which already see account-level flow, learn which questions about their participants public prices cannot answer, and why publishing class-level positioning, as some crypto venues already do, adds information that prices do not contain.
- *Risk managers and trading desks* get a realistic limit on what can be inferred about other traders or about crowding from market data, and a warning that common feedback measures computed from public data mostly reflect filtering and timing rather than true feedback (Supplement S13).
- *Researchers* who estimate models with several trader types get one reason why the parameters that separate the types are often imprecise, and a number that says how precise they can be before any estimation (Supplement S4).

What the data show. Two real reporting systems let us check the main prediction, that class records pin down what prices leave open. The U.S. Commodity Futures Trading Commission collects positions from large futures traders and publishes a weekly snapshot of them, aggregated by trader category. In a pre-specified test in eight futures markets (CME Bitcoin and Ether, the E-mini S&P 500, 10-year Treasury notes, the euro, the yen, crude oil and natural gas), these class positions pin down the persistence of the designated speculative class's positions to a range 9 to 71 times narrower than weekly prices allow, while prices alone leave it almost anywhere in the admissible range (Supplement S18). On Binance, in a pre-specified test on 30 markets never used before that matches the memory horizon of the two routes, the class data give the narrower interval in 29 of 29 evaluable markets, by a median factor of 7.3 (Supplement S19).

Robustness, a new failure and real noise. Rerunning the CFTC test with CME futures prices, without contract-roll weeks or with longer bootstrap blocks leaves its primary result unchanged. Read as weekly flows rather than positions, the same reports failed a second pre-specified test: they pin down how each class's flow responds to prices, but not how persistent that flow is. A third test asked whether positions carry a second, faster root that could be learning memory, as the model with inventory costs predicts; it failed too (2 of 8 markets), so these data cannot separate learning memory from the slow turnover of positions. Another pre-specified test gives synthetic groups real Bitcoin price noise instead of Gaussian noise and finds the predicted rates, a slope of -6.01 against -6 for two groups and -9.81 against -10 for three, while class-tagged requirements stay flat; at large sample sizes it uses a normal approximation to the real noise (Supplement S16). These comparisons are descriptive: a position series reveals its own persistence directly, while the price route must infer it. Several predictions failed, and Table 1 lists them next to the ones that passed. The data therefore show what prices cannot pin down. They do not show that this persistence is the learning memory of the model.

Two supplementary results. Neither is needed for the main results. Markets with the same eigenvalues, the same one-step volatility and the same long-run price variance can still have very different worst-case short-run losses, so watching eigenvalues or volatility cannot replace knowing who trades (Theorem S12.1). And standard feedback measures computed from public data mostly reflect filtering and clock alignment: two fee changes that moved trading volume by factors of 2.6 and 4.9 left measured feedback unchanged (Supplement S13).

Table 1: What the empirical tests found: passes and failures

Test	Prediction	Result
CFTC, 8 markets, H1	Class positions pin down a persistence that weekly prices leave open	Pass, 8 of 8
CFTC, H2	The designated speculative class trend-follows	Fail, 4 of 8 (three markets have the opposite sign)
CFTC, H3	That class has the less persistent positions	Pass, 6 of 8 (two indistinguishable)
CFTC flows (weekly changes), H1	Class flows pin down their own persistence (95% interval narrower than 0.5)	Fail, 3 of 8 (6 required); flow response nonzero in 6 of 8
CFTC positions, two-root test, H1	Positions carry a second, faster root (memory) distinct from the slow turnover root	Fail, 2 of 8 (6 required); where positive, the second root has a half-life under half a week
Real-noise injection, $K = 2$	Price-only requirement grows as $(\Delta\eta)^{-6}$; tagged requirement stays bounded	Pass: slope -6.01 (large-N points use a normal approximation); tagged max/min 2.8 (≤ 8 required)
Same test, $K = 3$ (secondary)	Price-only requirement grows as $(\Delta\eta)^{-10}$	Holds: slope -9.81
Binance, 30 fresh markets, (i)–(ii)	Class data restrict persistence; price-only intervals wide	Pass, 30 of 30 (one-hour price kernel)
Binance, same test, (iii)	Prices pin down the combined response	Fail, 5 of 30; the test failed its 80% criterion
Binance, 30 new markets	Horizon-matched: class interval narrower	Pass, 29 of 29 evaluable (median factor 7.3)
Binance, same test, secondary	Class persistence is a stable memory	Not supported: close to a random walk after bias correction
Binance, 105-day follow-up	Combined response far cheaper to learn than the separate rates	Fail, 10 of 30 (23 required)
Binance, 58-day follow-up	Tagging advantage grows with the sample; parametric shrinkage	Pass, 22 of 29; fail, 20 of 29 (22 required)
Binance, return forecasts	Class memory states improve 30-minute return forecasts	Fail ($\Delta R^2 = -1.09$ points, inconclusive)
Binance, state tracking	Class data track class positions better than prices	Pass, but explained by persistence of the target

Contribution. A large literature asks how transparency affects liquidity and trading costs (Pagano and Röell, 1996; Boehmer et al., 2005; Bessembinder et al., 2006). We ask a different question that market supervision now faces as trading passes to algorithms: what can an observer learn about *who* moves prices, and what is it worth to be told? The paper makes three contributions. First, it gives an identification result for supervision: a closed-loop price process reveals only the combined distribution of the trader groups' memories and weights, and nothing finer (Theorem 3.1), so questions about individual groups cannot be answered from public prices however much data accumulates. Second, it prices what it takes to go further, in the units a market participant or regulator faces, the number of experiments or observations, the quantity traded and the capital at risk, with matching upper and lower bounds where we can prove them (Sections 4 and 5); the cost explodes as $(\Delta\eta)^{-(4K-2)}$ when groups react over similar horizons, which turns the economic value of attribution into a number. Third, it values disclosure in the same units and shows when it changes a decision: class-tagged records keep the problem well conditioned (Proposition 6.1), sizing a market-wide limit needs only public prices, and deciding whether halting one group is enough needs the records (Proposition 6.4); published CFTC and Binance class positions pin down what prices leave open, with the failed predictions reported next to the passes (Section 6). The mathematical tools, moment matching, two-point testing and Prony-type estimation, are known from mixture estimation and signal processing; what is new is their closed-loop, passive and financial form (Supplement S4). Three of the central results, the identification theorem and the matching lower and upper bounds, are machine-checked in the Lean proof assistant, which is still rare in economics and finance.

How to read the paper. Every main result is stated formally and then explained in a short paragraph headed "In words", which says what it means for a trader or a supervisor; proofs are in the online supplement. Supplement S15 works through two desks that use Kalman filters, in numbers, and Table S7 maps each question to its answer. Readers who want the finance first can read this introduction, Section 2.1, the worked example and Section 6, and take the statistical results as given. Table S27 explains the main symbols.

What we do not claim. The identification results hold for the stated linear model and for the stated observation contracts, that is, for what the observer is assumed to see. The model treats price impact and the persistence of pricing errors as known primitives rather than solving a dealer equilibrium, and it assumes the observer knows the fundamental value; real data meet neither assumption. They are not general limits on surveillance with participant-level records, which can be far more informative (Propositions 4.4 and 6.1). The matching

upper bound holds on a class with separated rescaled rates and weights bounded away from zero. Capital figures depend on the observation-noise calibration, which we estimate rather than assume. The empirical results are reduced-form diagnostics; they do not identify structural learning parameters. Our pre-specified real-data test of the identification boundary failed its criterion on one of its three parts: on 42 days of data the merged response was precisely estimated in only 5 of 30 markets (Supplement S19). Binance's class ratios are positions, not tagged order flow, so they are the closest public analogue to the disclosure of Proposition 6.1, not an instance of it. We hold the algorithms' memories and intensities fixed across disclosure regimes; disclosure that reveals positions may lead informed desks to trade less or to split orders across accounts, which we do not model.

1.1 Relation to the literature

Three literatures meet here. In finance, learning algorithms and AI agents in markets are studied by simulating or modelling them (Colliard et al., 2022; Dou et al., 2025); traders who differ in horizon or speed are studied in equilibrium (Cespa and Vives, 2015; Biais et al., 2015; Crouzet et al., 2020); the exponentially fading beliefs we use are the extrapolative beliefs of Barberis et al. (2015, 2018); and heterogeneous-agent models estimated from prices often leave the type-separating parameters imprecise (Brock and Hommes, 1998; Amilon, 2008; Kukacka and Barunik, 2017), for which our results give one reason. Price formation follows the linear-impact tradition (Kyle, 1985; Glosten and Milgrom, 1985; Hasbrouck, 1991) and the transient-impact literature (Obizhaeva and Wang, 2013; Gatheral, 2010; Alfonsi et al., 2012); trader types have been inferred from public data, as in the probability of informed trading (Easley et al., 1996), and from account data (Kirilenko et al., 2017). Learning by trading has an equilibrium counterpart in price experimentation (Leach and Madhavan, 1993), and execution that learns impact while trading is studied by Moallemi et al. (2008) and Park and Van Roy (2015). Technically, the closest work is closed-loop identification (Ljung, 1999; Forssell and Ljung, 1999), weak identification (Andrews and Cheng, 2012), and super-resolution and mixture estimation (Moitra, 2015; Heinrich and Kahn, 2018; Wu and Yang, 2020; Batenkov et al., 2021; Katz et al., 2023). We do not claim the K-group exponent as new: in mixture language it is the square of the condition number for K clustered nodes (Heinrich and Kahn, 2018; Wu and Yang, 2020), and Prony's method attains it (Katz et al., 2023); the δ^{-4} law for one against two close components is classical in resolution analysis (Shahram and Milanfar, 2005); logarithmic rates are known for mixtures of exponentials (Rebafka and Roueff, 2015); and labelled samples are known to be more informative than unlabelled ones (Castelli and Cover, 1996). What is new is the closed-loop form (the memories are zeros,

not poles, of the observed transfer function, so they must first be extracted from the loop), the passive equivalence, the translation of sample sizes into executed quantity, capital and money, and the value of class records, including a decision in which attribution changes the right action. Supplement S4 gives the full comparison, with tables of the closest results.

1.2 Outline

Section 2 sets out the model and Section 3 the identification boundary. Section 4 turns memory resolution into executed quantity, capital and money for a known pair, and Section 5 treats unknown mixtures, general memory and passive data. Section 6 states what disclosure buys and when it changes a supervisor's decision, and tests it on the CFTC's mandated trader-class reports and on Binance's class ratios. Section 7 draws the implications for supervision and Section 8 concludes. The online supplement contains the proofs, the extended literature review and the map of results, the discussion of the assumptions, general memory, the real-noise test, the trader-class tests in full, further results and robustness checks, the hypothetical dollar illustration, a finite-horizon decision example, the spectral blind spot, the measurement of reduced-form feedback, the formal-verification statement and the reproducibility map.

2 Model and motivating examples

The model in words. The market has one price and K groups of traders. Each period three things happen. First, the price moves: it drifts back toward its fundamental value, and it is pushed by the groups' trades, by any test trade and by news. Second, each group updates its view by mixing the newest price into its running average, with weight η_i on the newest price and $1-\eta_i$ on its old view. Third, each group trades in proportion to its view, with intensity q_i. A group with a large η_i has a short memory and reacts quickly, like a short-horizon trend follower; a group with a small η_i has a long memory and reacts slowly. The equations below write this down.

Time is discrete. Let p_t be the price, p^* the fundamental value and $x_t = p_t - p^*$ the pricing error. A reduced-form dealer sector absorbs net order flow with price impact $\kappa > 0$, an imposed primitive rather than an equilibrium quote, and arbitrage pulls the error toward zero with persistence $a \in [0, 1)$:

$$x_t = a x_{t-1} + \kappa \sum_{i=1}^{K} q_i y_{i,t-1} + \kappa v_t + w_t, \tag{1}$$

where v_t is an executed probe quantity and w_t a price innovation. Cohort i holds a state $y_{i,t}$ with exponential memory and trades $q_i y_{i,t}$:

$$y_{i,t} = (1-\eta_i)\, y_{i,t-1} + \eta_i x_t, \qquad \eta_i \in (0,1). \tag{2}$$

The ordering is *price first*: the price responds to the probe, then cohorts update. Write $r_i = 1-\eta_i$ for the memory rate and $w_i = q_i\eta_i$ for the merged weight. The feedback kernel and its transform are

$$k_m = \sum_i w_i r_i^m, \qquad G(z) = \sum_i \frac{w_i z}{z - r_i}, \tag{3}$$

and the transfer function from probes to prices, $H(z) = \sum_{j\geq 0} h_j z^{-j}$, is

$$H(z) = \frac{\kappa z}{z - a - \kappa G(z)} = \frac{\kappa z D(z)}{(z-a)D(z) - \kappa N(z)}, \qquad G = \frac{N}{D}, \quad D(z) = \prod_i (z - r_i), \tag{4}$$

with impulse response $h_0 = H(\infty) = \kappa, h_1, h_2, \ldots$; in particular $H(1) = \kappa/(1-S) = \sum_j h_j$. Stacking $z_t = (x_t, y_{1,t}, \ldots, y_{K,t})^\top$ gives $z_t = Jz_{t-1} + bv_t$, where, because cohorts update on the same period's price, the probe enters every state on impact through

$$b = \kappa\,(1, \eta_1, \ldots, \eta_K)^\top, \tag{5}$$

and

$$J = \begin{pmatrix} a & \kappa q_1 & \cdots & \kappa q_K \\ \eta_1 a & 1-\eta_1+\eta_1\kappa q_1 & \cdots & \eta_1\kappa q_K \\ \vdots & & \ddots & \vdots \\ \eta_K a & \eta_K\kappa q_1 & \cdots & 1-\eta_K+\eta_K\kappa q_K \end{pmatrix}, \qquad S \equiv a + \kappa\sum_i q_i. \tag{6}$$

In the *positive economy* ($q_i \geq 0$), J is entrywise nonnegative, its price row sums to S, and the economy is Schur stable iff $S < 1$.

Lemma 2.1 (Two motivating examples: filtering and learning)**.** *(i)* Kalman memory. *A desk that tracks a latent random-walk trend $v_{t+1} = v_t + \zeta_{t+1}$, $\zeta \sim N(0, \sigma_\zeta^2)$, from signals $r_t = v_t + \nu_t$, $\nu \sim N(0, \sigma_\nu^2)$, using the steady-state Kalman filter, updates as $\hat{v}_{t+1} = (1-\eta)\hat{v}_t + \eta r_t$ with*

$$\eta = \frac{-\sigma_\zeta^2 + \sqrt{\sigma_\zeta^4 + 4\sigma_\zeta^2\sigma_\nu^2}}{2\sigma_\nu^2} \in (0,1). \tag{7}$$

(ii) Gradient memory. *Online gradient descent with L_2 regularization $\gamma \in (0,1)$, step size*

$\alpha > 0$ and loss $\frac{1}{2}H(x_t - \theta_t)^2$ gives $\theta_{t+1} = (1-\eta)\theta_t + \eta q x_t$ with $\eta = \gamma + \alpha H \in (0,1)$ and $q = \alpha H/\eta$. A tabular value update with learning rate η at a fixed state has the same form; general reinforcement-learning policies need not.

Remark 2.2 (Why exponential memory). Lemma 2.1 shows that two different design philosophies, Bayesian filtering and regularized online learning, produce the same observable reduced-form belief state $y_{i,t}$. When desk i chooses order flow $u_{i,t}$ subject to $I_{i,t} = (1-\delta_{\text{inv}})I_{i,t-1} + u_{i,t}$ and an inventory cap $|I_{i,t}| \le \bar{I}_i$ to maximize expected continuation return $\beta_i y_{i,t} u_{i,t}$ net of quadratic execution cost $\frac{1}{2}\lambda_{\text{tc},i} u_{i,t}^2$ and quadratic inventory holding cost $\frac{1}{2}\gamma_{\text{inv},i} I_{i,t}^2$, its interior first-order condition gives $u_{i,t}^* = q_i y_{i,t} - \psi_{\text{inv},i} I_{i,t-1}$ with market-response weight $q_i = \beta_i/(\lambda_{\text{tc},i}+\gamma_{\text{inv},i})$ and inventory mean-reversion $\psi_{\text{inv},i} = \gamma_{\text{inv},i}(1-\delta_{\text{inv}})/(\lambda_{\text{tc},i}+\gamma_{\text{inv},i})$, which shrinks toward zero as trading costs or inventory penalties tighten and destabilizes the market when the closed-loop spectral radius reaches one. This filtering-and-control recursion specifies an individual agent's decision rule, not a complete economic equilibrium (dealer–trader equilibrium is analysed in the companion paper).

Two further simplifications matter for reading the results: the observer is taken to know the fundamental value p^*, so the pricing error x_t is observed, and the dealer sector is a fixed linear impact κ rather than an equilibrium quote. The paragraph before Supplement S18 states what the data tests lose when neither holds. Assuming that the observer knows p^* gives it more information than any real observer has, so the lower bounds hold a fortiori when p^* is unknown; the upper bounds use it.

The model applies these rules to the pricing error x_t: a desk that cannot separate fundamental news from transitory deviations treats recent deviations as a signal of a persistent trend, which is the exponential extrapolation of Barberis et al. (2015, 2018). With $q_i > 0$ such a desk buys after rises, as trend followers do, and nothing in the model says this is profitable; the lemma shows only that the update has the exponential form. In these two illustrative families of algorithms the memory rate is a known function of a tuning parameter, the desk's signal-to-noise ratio or its regularization. That is the kind of private information a supervisor would like to recover, although real trading algorithms need not belong to either family. Section 3 shows why prices reveal it only in merged form.

Lemma 2.3 (Positive price-response geometry)**.** *In the positive, stable economy the impulse response is a nonnegative mixture of decaying exponentials,*

$$h_j = \sum_{\ell} \omega_\ell \lambda_\ell^j, \qquad \omega_\ell \ge 0, \quad \sum_{\ell} \omega_\ell = \kappa, \quad 0 \le \lambda_\ell < 1.$$

It is completely monotone and nonincreasing; its peak is the direct impact κ, even though J is non-normal. In price/memory coordinates the price-only gain is $\sup_t \|e_x^\top J^t\|_2 = 1$.

As an illustration of the lemma (not part of its proof), $(-1)^n \Delta^n h_j \geq 0$ for $n \leq 4$ held in all 892 stable random positive economies tested. Lemma 2.3 matters twice: it makes probing costs positive (Section 4), and it shows that in this economy transient amplification of the price response cannot arise from cohort feedback alone. The companion paper locates amplification in order-book depth instead.

Assumptions. Supplement S14 discusses the six modelling choices behind the results: linearity, price-first timing within a period, exponential memory (relaxed in Supplement S17), algorithms that react to being probed, known primitives, and the observation contracts. Unknown primitives can only make separation harder, so the lower bounds remain valid; the upper bounds as stated assume the primitives are known.

2.1 The observer, its options and its costs

The results are stated from the side of one observer outside the trader groups: a supervisor, an exchange's surveillance unit or a large desk. Its *target* is attribution, recovering the merged weight and memory rate of each group to a stated accuracy, or deciding between two candidate populations at a stated error. Before it observes, it knows the price persistence a, the impact κ and the law of the observation noise, and, for the K-group results, the number of groups; it does not know their rates or weights. These are the conditions under which the lower bounds are statements about the data rather than about the observer's ignorance; unknown primitives can only enlarge the set of candidate populations.

The observer has three options, and each result prices one of them.

(a) *Wait.* Record stationary prices driven by natural innovations, without trading. The cost is time: the number of observations needed (Theorems 5.5 and 5.6).

(b) *Probe.* Execute a sequence v_t of trades against the groups and record the price response. Probes execute at the average price of a linear book after the uncontrolled state update; the prober bears the execution loss $\frac{1}{2} v^\top T_h v$ plus fees (Lemma 4.2) and must fund margin throughout. The counted resource is the number of independent experiments, each starting from the reset state, or the executed quantity and capital of a single experiment (Theorems 4.1–S9.2).

(c) *Acquire disclosure.* Receive each class's flow or position, observed with noise, at an operating cost per supervisory cycle and a compliance cost per market (Proposition 6.1, Theorem 6.2, Proposition 6.3).

Within each period the order is: the probe (if any) executes and the price updates, the groups update their states on the new price, and the observer records the price with noise. The

groups do not change their rules in response to being observed; Supplement S14 discusses what happens if they do. Probing experiments start from the reset state. A probe sent into a running market adds the free response of the current state, which by linearity superposes with the forced response; if the state is unknown it acts as a nuisance term, which can only make separation harder, so the lower bounds still hold, while the upper bounds as stated need the reset state or a known one. Independent resets are a benchmark experiment that no outside participant can run in a live market; we use them because they make the information content of the data exact, and the economic comparisons say whenever they rely on them. Regulators do not, in practice, trade to learn; the probing option describes an outside participant, or prices the information a supervisor would otherwise have to buy by trading.

Three money quantities appear and are kept apart: the *capital required*, the equity an account must hold so that its margin constraint is never breached while it probes (Theorem 4.3 and Lemma 4.5); the *round-trip trading loss*, the execution loss of a probe that ends flat, which is what the account actually spends (Theorems S9.2, S9.3 and S9.4); and the *annualised loss* of renewing that probe weekly, used for the comparison with audit-trail budgets (Supplement S8). The lower bounds hold for the stated probe classes, deterministic probes and, in Theorem S9.3, non-anticipating sequential probes; they are not lower bounds on information obtained through other channels, which is the point of Section 6. Two rates recur and measure different things: δ^{-4} is the capital (and the average-power horizon) for a *known* symmetric pair whose rates differ by 2δ, while $(\Delta\eta)^{-(4K-2)}$, which is $(\Delta\eta)^{-6}$ for $K = 2$, is the number of experiments for K *unknown* groups with spacing $\Delta\eta$.

3 The identification boundary

A central question in algorithmic market surveillance is whether an observer of executions and prices can identify the individual strategies active in the market.

Theorem 3.1 (Merged rate–weight equivalence)**.** *Let $\kappa > 0$ and $a \in [0, 1)$ be known. Use zero initial states and a common probe contract, and observe the modeled price state x_t.*

(i) Equivalence class. *Two economies $\mathcal{E}^A = \{(q_i^A, \eta_i^A)\}$ and $\mathcal{E}^B = \{(q_j^B, \eta_j^B)\}$ produce identical price responses for every probe sequence if and only if their merged measures coincide:*

$$\sum_{i:\eta_i^A=\eta} q_i^A \eta \;=\; \sum_{j:\eta_j^B=\eta} q_j^B \eta \qquad \forall \eta \in (0, 1]. \tag{8}$$

Write $a' = a + \kappa M(\{0\})$, so that cohorts with $\eta = 1$ (rate 0) act as part of the anchor.

For $a' \neq 0$ the minimal realization has dimension $1 + |\mathrm{supp}(M) \cap (0,1)|$, counting distinct positive rates with nonzero merged weight after signed cancellation; for $a' = 0$ the price carries no state of its own and the dimension is $|\mathrm{supp}(M) \cap (0,1)|$. A cohort with $\eta = 1$ merges into the anchor a without adding a pole.

(ii) Separation complexity under average power. *Consider testing a single cohort with rate $\bar{\eta}$ against two cohorts with rates $\bar{\eta} \pm \delta$ and the same total weight, on a compact family that is uniformly Schur stable with all rates in $[\epsilon, 1-\epsilon]$ for some $\epsilon > 0$. Observe $\tilde{x}_t = x_t + \nu_t$ with $\nu_t \sim N(0, \sigma_\nu^2)$ and use deterministic probes with $\frac{1}{T}\sum_{t<T} v_t^2 \leq P$. Any test with worst-case error at most $\alpha \in (0, 1/4)$ needs*

$$T \;\geq\; \frac{4\sigma_\nu^2(1-2\alpha)^2}{C_{\mathcal{F}}^2 P\, \delta^4} = \Omega(\delta^{-4}), \tag{9}$$

where $C_{\mathcal{F}}$ depends only on the family, through ϵ, the bound on $|H|$ implied by uniform stability, and the total weight q.

In words. If two trading desks update on prices with the same memory, no amount of price data can tell them apart or say how much each trades: prices reveal only how much trading happens at each memory speed. Desks with different memories can be told apart in principle, and the rest of the paper prices how hard that is.

The memory rates are poles of G and *zeros* of H, not closed-loop poles. This has a practical consequence we verify in Section 5: pole-fitting methods applied directly to the price impulse response recover closed-loop eigenvalues (0.4256 and 0.8224 for $K = 1$, $\eta = 0.3$, $a = 0.5$, $\kappa = 0.8$, $q = 0.2$) and miss the memory rate 0.70.

Remark 3.2 (Scope). The result concerns the stated model and observation contract. Cohorts sharing an exact rate enter prices through their merged weight; this does not determine identities or wealth allocations. Nearby rates can be distinguishable with enough information. The δ^{-4} statement in (ii) is a lower bound on a declared family, with family-dependent constants. Individual algorithms may choose their rates strategically, but any unilateral change maps into the same observer-facing merged measure, which reinforces rather than circumvents the boundary.

4 Probing resources, memory resolution and money

Observation length alone cannot buy identification when the lifetime quantity of controlled executions is fixed. This section connects the merged-memory experiment to a financially

specified input/output contract. Throughout, Φ is the standard normal distribution function.

4.1 Quantity

Theorem 4.1 (Sharp probing-resource orders for a known symmetric pair)**.** *Fix $0 \le a < 1$, $\kappa, q > 0$, $S = a + \kappa q < 1$ and $\eta \in (0,1)$. Compare a single cohort of weight q and rate η with two cohorts of weight $q/2$ and rates $\eta \pm \delta$, both in $[\epsilon, 1-\epsilon]$. All parameters are known. The price receives executed quantity through κv_t, starts from zero, has no process noise, and is observed with independent $N(0,\sigma^2)$ noise. Probes are deterministic and common to both alternatives. Let $d_T = (h^\delta_j - h^0_j)_{j<T}$ and $K_* = 2\kappa^2 q/[\epsilon^3(1-S)^2]$.*

(i) For every probe with $\sum_t |v_t| \le V$,

$$D_{\mathrm{KL}}(P^v_0 \| P^v_\delta) \le \frac{K_*^2 V^2 \delta^4}{2\sigma^2}, \qquad e_{\mathrm{worst}} \ge \frac{1}{2}\Big(1 - \frac{K_* V \delta^2}{2\sigma}\Big)_+. \tag{10}$$

(ii) The optimal worst-case error over these probes and all classifiers is exactly

$$e^*_T(V) = \Phi\left(-\frac{V\|d_T\|_2}{2\sigma}\right), \tag{11}$$

attained by a pulse at the first input and the likelihood-ratio test. Waiting adds information but $\|d_T\|_2 \uparrow \|d_\infty\|_2 < \infty$, so it cannot make the error arbitrarily small at fixed V. Error at most $\alpha < 1/2$ requires $V \ge 2\sigma(1-2\alpha)/(K_*\delta^2)$ independently of T.

(iii) The first two taps coincide and $d_2 = -\kappa^2 q \delta^2$. A single pulse of size

$$A = \frac{2\sigma\Phi^{-1}(1-\alpha)}{\kappa^2 q \delta^2} \tag{12}$$

and the third observation achieve error exactly α. The necessary and sufficient quantity order is δ^{-2}.

(iv) Under average power $\sum_{t<T} v_t^2 \le PT$, the necessary horizon order is δ^{-4}, attained by the pulse whenever $T \ge \max\{3, \lceil A^2/P \rceil\}$.

(v) For a fixed monetary price exposure $\lambda \ne 0$, additive reserve L and cost envelope $c\sum|v_t|$, the sharp robust quantity allowance under budget $B > L$ is $V_B = (B-L)/(|\lambda|\kappa + c)$.

In words. To tell one desk from two desks whose memories differ slightly, the observer must trade against them, and the quantity it must trade grows like $1/\delta^2$: halving the difference

quadruples the trading needed. Waiting longer after a fixed trade does not help, because the trade's echo in the price dies out.

The identity in (iii) follows by direct substitution, and code checks it at $\delta \in \{0.02, 0.05, 0.10\}$: the difference is -1.2×10^{-4}, -7.5×10^{-4} and -3.0×10^{-3} at $\kappa = 1$, $q = 0.3$, and -3.2×10^{-4} at the calibration $\kappa = 0.8$, $q = 0.2$, $\delta = 0.05$. The theorem does not solve estimation of an unknown mixture (Section 5).

A Monte Carlo check of the exact error. Formula (11) is exact for the likelihood-ratio test. We simulated 200,000 noisy paths per hypothesis at the calibration ($\sigma = 0.01$, $\delta = 0.05$) for several pulse sizes and horizons (Table S28); simulated errors match the formula within Monte Carlo standard error under both hypotheses.

Figure S6 shows why the problem is hard. The two economies have the same direct impact and the same first-lag response; they first differ at lag two, by an amount proportional to δ^2, and the whole difference sequence scales with δ^2. A test therefore needs the signal-to-noise ratio $V\|d\|/\sigma$ to grow like δ^{-2}, which is the quantity order, and because the adverse mark of a probe grows with the square of its size, capital grows like δ^{-4}.

4.2 Execution cost

A probe changes inventory as well as price. We therefore specify cash and execution timing. Price has units of currency per asset, and κ currency per squared quantity.

Lemma 4.2 (Fair execution)**.** *For a quote schedule refreshed after the uncontrolled state update, execution at the average price of a linear book costs, relative to the constant fundamental mark,*

$$C(v) = \tfrac{1}{2} v^\top T_h v + c\|v\|_1, \qquad (T_h)_{ij} = h_{|i-j|}. \tag{13}$$

In the positive stable economy T_h is positive definite, so no nonzero round trip produces a deterministic gain.

We verify the ledger identity (13) against an explicit cash account to 5.7×10^{-13} on 1,000 random probes, and the smallest eigenvalue of T_h over 1,000 random stable positive economies is 0.141. A direct-only proxy $\sum \frac{1}{2}\kappa v_t^2$ omits cross-time impact: for $v = (+1, -1)$ it gives 0.80 against a true Toeplitz loss of $h_0 - h_1 = 0.3488$.

4.3 Capital supplied by a counterparty

Theorem 4.3 (Sharp robust quantity cap from cash and margin)**.** *Use the positive, stable, reset economy with constant fundamental price $P_* > 0$ and no process noise. A designated*

dealer fills only nonnegative probes $v_t \geq 0$ at the average price $P^0_{t+1} + \kappa v_t/2$ of a linear schedule, holds inventory $I_{t+1} = I_t - v_t$ and cash $K_{t+1} = K_t + v_t(P^0_{t+1} + \kappa v_t/2) - cv_t$, starts with equity $B > 0$, and must satisfy $\mu|I_t|P_t \leq E_t$ at every time. For the robust contract admitting every probe with $\sum_t v_t \leq V$, the largest allowance guaranteeing margin is

$$V_B = \frac{2B}{\beta + \sqrt{\beta^2 + 4(\mu + \frac{1}{2})\kappa B}}, \qquad \beta = c + \mu P_*, \tag{14}$$

equivalently $(\mu + \frac{1}{2})\kappa V^2 + (c + \mu P_*)V \leq B$. The cap holds across all positive cohort representations with $S < 1$, without resolving individual rates. An observer that receives only noisy prices and executed quantities has best error $\Phi[-V_B\|d_T\|_2/(2\sigma)]$ within this allowance; fixed error $\alpha < 1/2$ requires capital of order at least δ^{-4}, and a single positive pulse attains that order.

In words. A probe moves the price against the account that makes it, so the margin it needs grows with the square of the probe. Because the probe grows like $1/\delta^2$, the capital grows like $1/\delta^4$: halving the difference multiplies the capital needed by 16.

Even with zero initial inventory, the probe creates a quadratic adverse mark that a fixed initial price exposure would miss. Figure S7 illustrates the experiment.

Proposition 4.4 (Exact financial disclosure changes the experiment)**.** *In this noiseless-state account, exact disclosure of the third equity mark after a single positive pulse $0 < A \leq V_B$ distinguishes the known pair with zero error for every $\delta > 0$.*

In words. If the observer can see an account's exact equity, a single test trade settles the question, however close the two candidate markets are. The hard part of the problem is therefore not the market itself but what the observer is allowed to see.

Exact marked equity is extra information, not a free observation assumed by the price-only theorem. Real account marks have their own precision: with reports rounded to 0.01, a guaranteed change requires a larger trade (Section 6).

4.4 Capital paid by the probe trader

Lemma 4.5 (Time-domain Toeplitz coercivity: capital order for funded probes)**.** *Use the same economy and execution law. The statistical observer receives only noisy prices and executed quantities. The probe trader starts with cash $B > 0$ and zero inventory; a finite signed sequence v changes inventory by $+v_t$ and cash by $K_{t+1} = K_t - v_t(P^0_{t+1} + \kappa v_t/2) - c|v_t|$,*

after which trading stops and the trader must remain solvent as the price returns to P_*. Put

$$r = 1 - \epsilon(1 - S) < 1, \qquad m = \kappa \frac{1 - r}{1 + r} > 0, \qquad K_* = \frac{2\kappa^2 q}{\epsilon^3 (1 - S)^2}.$$

Every solvent strategy satisfies $D_{\mathrm{KL}}(P_0^v \| P_\delta^v) \le K_*^2 B \delta^4 / (\sigma^2 m)$, so error at most $\alpha < 1/2$ requires

$$B \;\ge\; \frac{2\sigma^2 m (1 - 2\alpha)^2}{K_*^2 \delta^4}, \tag{15}$$

without a quantity cap and at any horizon. Conversely, for any margin fraction $0 < \mu \le 1$, the pulse A in (12) achieves error α and remains margin-solvent at every later mark with capital $B_A = \frac{\kappa}{2} A^2 + (c + \mu P_*) A$. Necessary and sufficient capital orders are δ^{-4}.

Lemma 4.5 is an order statement: an elementary time-domain coercivity argument for Toeplitz forms that establishes the δ^{-4} rate before the exact frequency-domain optimization of Supplement S9. Its explicit constant is very conservative, because it scales as $m/K_*^2 \propto \epsilon^7 (1 - S)^5$: at the calibrated parameters it is 4.9×10^{-4} par units at $\delta = 0.05$ with $\epsilon = 0.35$, five orders of magnitude below any feasible design. We therefore state it as a lemma; Supplement S9 computes the sharp constant, within 5% of a feasible design at every separation.

A worked example with two Kalman desks that differ only in their signal-to-noise ratios (Supplement S15) puts these orders into numbers.

5 Unknown mixtures and passive data

Theorem 4.1 concerns a *known* pair. A supervisor faces an unknown number of cohorts with unknown rates and weights. This section asks how fast the cost of recovering them grows.

5.1 Why naive pole fitting fails

Because the memory rates are zeros of H (Theorem 3.1), extracting poles directly from the price impulse response by matrix-pencil or Prony methods targets closed-loop eigenvalues. A four-step inversion repairs this: (1) decouple $g_t = h_{t+1} - a h_t$; (2) form a Hankel matrix of g and compute a minimal realization by Kung's eigensystem realization; (3) compute its transmission zeros, which are the roots of the denominator of G; (4) project reconstructed flow onto the recovered memory filters to obtain the weights (Supplement S3.3). Without noise this recovers $K = 0$, $K = 1$ and well-separated two-cohort systems exactly (Supplement S3.3); with noise, per-cohort weights become unreliable once rates are close. The

reason is information-theoretic, as the rest of this section shows.

5.2 A global two-point bound for any estimator

The Cramér–Rao benchmark of Proposition S7.1 applies only to regular unbiased estimators. Could biased, ridge, shrinkage or Bayesian estimators escape the $(\Delta\eta)^{-6}$ scaling? No.

Theorem 5.1 (Global two-point bound and matching upper bound, $K = 2$)**.** *In this theorem w_k denotes cohort k's trading intensity q_k, not the merged weight $q_k\eta_k$ of Section 2, and $\Delta\eta$ is the actual spacing of the two rates. Fix total intensity $q = 0.20$, mean rate $\bar{\eta} = 0.40$, target separation $\tau = 0.01$, and $\Delta\eta \in (0, 8/11)$ (so that $\bar{\eta} - p_+\Delta\eta > 0$). Let $p_\mp = \frac{1}{2} \mp \tau/q$ and*

$$\theta_0 = (qp_-, qp_+, \bar{\eta} - p_+\Delta\eta, \bar{\eta} + p_-\Delta\eta), \qquad \theta_1 = (qp_+, qp_-, \bar{\eta} - p_-\Delta\eta, \bar{\eta} + p_+\Delta\eta). \tag{16}$$

(i) Label cohorts in increasing order of rate, so that weight vectors are compared rate by rate (as unordered sets the two weight vectors coincide). Both economies have spacing $\Delta\eta$, weight separation $\|w(\theta_1) - w(\theta_0)\|_\infty = 2\tau$ in this labelling, and equal rate moments $M_m = \sum_k w_k\eta_k^m$ for $m \in \{0, 1, 2\}$, while

$$M_3(\theta_1) - M_3(\theta_0) = 4\tau p_- p_+ (\Delta\eta)^3. \tag{17}$$

Hence $\|h(\theta_1) - h(\theta_0)\|_2 = \Theta(\tau(\Delta\eta)^3)$.

(ii) For any measurable estimator (including biased, ridge, James–Stein or MAP) that reports weights in the same rate order, the supremum being over the class of two-cohort economies with total weight q, mean rate $\bar{\eta}$ and spacing $\Delta\eta$,

$$\sup_\theta P_\theta(\|\hat{w} - w(\theta)\|_\infty \geq \tau) \;\geq\; 1 - \Phi\Big(\frac{\sqrt{N}\|h(\theta_1) - h(\theta_0)\|_2}{2\sigma_h}\Big), \tag{18}$$

so failure probability at most 0.05 requires $N \geq 4\sigma_h^2 z_{0.95}^2/\|h(\theta_1) - h(\theta_0)\|_2^2 = \Omega((\Delta\eta)^{-6})$. For $\Delta\eta \geq 2\tau q/(q + 2\tau)$ (≈ 0.018 at the stated values) the same bound holds without any labelling convention, for the permutation-invariant loss

$$d_\infty(\hat{\theta}, \theta) = \min_{\pi \in S_2} \max_k \big(|\hat{w}_{\pi(k)} - w_k| + |\hat{\eta}_{\pi(k)} - \eta_k|\big),$$

because then $d_\infty(\theta_0, \theta_1) = \min\{2\tau(1 + \Delta\eta/q),\, (1 + 2\tau/q)\Delta\eta\} \geq 2\tau$. For smaller spacings the relevant permutation-invariant statement is the Wasserstein bound of Theorem 5.3.

(iii) Conversely, the matching upper bound $N = O((\Delta\eta)^{-6})$ on rich separated two-cohort

classes is Theorem 5.4 with $K = 2$. On the separated class with weights and rates bounded away from the boundary, Toeplitz deconvolution followed by a Prony–Hankel root step has Hankel determinant $g_1 g_3 - g_2^2 = w_1 w_2 \eta_1 \eta_2 r_1 r_2 (\Delta\eta)^2 > 0$ for the deconvolved kernel $g_m = \sum_i w_i \eta_i r_i^m$, and perturbation analysis gives the same order.

In words. For two unknown groups, no method can tell the two candidate markets apart reliably with fewer than a number of experiments of order $(\Delta\eta)^{-6}$, and a practical method (Prony's method, Theorem 5.4) achieves this order. Halving the spacing multiplies the data needed by 64.

We recomputed the two-point quantities exactly (Table S29): the moment gap equals $4\tau p_- p_+ (\Delta\eta)^3$ to machine precision, and the Le Cam requirement at $\Delta\eta = 0.10$ is 5,927.6 resets. The worst-case failure rates of four re-implemented estimators all lie above the lower bound (Table S29).

Remark 5.2 (The hard directions carry no risk). The two economies of Theorem 5.1 have the same total weight $\sum_i q_i = q$, so their summed price response $\sum_t h_t = \kappa/(1 - a - \kappa \sum_i q_i)$ is identical, 2.353 at the calibration of Table S29: the directions in which prices are least informative about the cohorts are directions in which cumulative price risk does not move. The summed response is a linear functional of the response and is estimated at the parametric rate whatever the spacing of the memories; the companion paper turns this into a certification theorem, a minimax-optimal pre-trade guard and a decision value of disclosure (Niroula et al., 2026). Identification and safety ask different questions of the same data.

5.3 Many unknown cohorts

Theorem 5.3 (K-cohort two-point lower bound)**.** *Fix $K \geq 2$, a center $r_c \in (0, 1)$, total merged weight $W > 0$ and $0 < \Delta\eta < \min(r_c, 1 - r_c)$, so that all rates lie in $(0, 1)$, and assume $a + \kappa W/(1 - r_c - \Delta\eta) < 1$, so that every candidate economy is Schur stable. Each reset experiment observes $T \geq 2K$ noisy taps ($T = \infty$ allowed); the bound below uses the full response norm, which dominates any finite prefix. There exist two K-cohort economies with rates in $r_c + \Delta\eta[-1, 1]$, positive weights, and merged measures whose moments of order $0, \ldots, 2K-2$ coincide while the moment of order $2K-1$ differs by a constant times $(\Delta\eta)^{2K-1}$. Their impulse responses differ by $\|h^A - h^B\|_2 \asymp (\Delta\eta)^{2K-1}$, so recovering the weights from N independent noisy reset experiments with worst-case error probability below $1/4$ requires*

$$N \;\geq\; c\,\sigma^2 (\Delta\eta)^{-(4K-2)}.$$

The same requirement holds for the permutation-invariant loss $W_1(\hat{M}, M)$ between estimated and true merged measures at accuracy $c_K W \Delta\eta$, where $2c_K > 0$ is the Wasserstein distance between the two normalized quadrature rules, so no relabelling of atoms escapes it.

The two losses have different targets. The weights of the two economies differ by a constant that does not shrink with $\Delta\eta$, so the first statement is about weights at fixed accuracy, which Theorem 5.4 matches; the Wasserstein statement is about the memory distribution at accuracy proportional to $\Delta\eta$, which Theorem S10.1 matches.

The construction uses the K-point Gauss and Gauss–Radau rules of the same weight function: both are positive, have K atoms and integrate polynomials of degree $2K-2$ exactly. For $K = 2$ it reproduces the order of Theorem 5.1. Table S30 lists the two rules: their low moments agree to machine precision and their moment of order $2K-1$ differs by 0.222, 0.0533 and 0.0131 for $K = 2, 3, 4$. The shrinking gap explains why constants deteriorate with K on top of the exponent.

What counts as an experiment. In Theorem 5.3 an experiment is a unit impulse from the reset state whose response is observed with noise of variance σ^2 per tap. The exponent is the same for any probe design of bounded energy. For any probe v the mean difference between the two economies is at most $\sup_\omega |\Delta\hat{H}(e^{i\omega})|\,\|v\|_2$, and $\sup_\omega |\Delta\hat{H}| = O((\Delta\eta)^{2K-1})$ because $\Delta\hat{H} = z^{-1}\hat{H}_A\hat{H}_B\,\Delta G$ with both responses bounded on the circle. Experiments with probe energy E each therefore need $NE \gtrsim \sigma^2(\Delta\eta)^{-(4K-2)}$: a larger probe buys the same information with fewer experiments, and Supplement S9 prices the capital that a larger probe requires. The bound does not cover information obtained without trading against the groups, such as class records (Section 6).

Table S31 and Figure S8 confirm the order numerically.

Theorem 5.4 (Matching upper bound for every K)**.** *Fix $K \geq 2$, a compact set $\mathcal{S} \subset \{s \in [-1,1]^K : s_1 < \cdots < s_K\}$ of ordered rescaled rates with $\min_{i\neq j} |s_i - s_j| \geq \gamma > 0$, and merged weights $w_i \in [\omega_{\min}, \omega_{\max}]$ with $\omega_{\min} > 0$. Consider the class of K-cohort economies with rates $r_c + \Delta\eta\, s$, $s \in \mathcal{S}$, and N independent reset experiments whose responses carry Gaussian noise with variance σ^2. There are constants $c_0, C > 0$ depending only on $(K, \mathcal{S}, \gamma, \omega_{\min}, \omega_{\max}, a, \kappa, r_c)$ such that, whenever $\epsilon_N \equiv \sigma N^{-1/2}(\Delta\eta)^{-(2K-1)} \leq c_0/\sqrt{\log(4K/\beta)}$, Prony's method applied to the first $2K$ estimated kernel values, with its output atoms listed in increasing order of rate, recovers the weights with*

$$\max_i |\hat{w}_i - w_i| \;\leq\; C\sqrt{\log(4K/\beta)}\;\epsilon_N$$

with probability at least $1-\beta$. Hence $N = C'\sigma^2(\Delta\eta)^{-(4K-2)}$ experiments suffice for any fixed

accuracy and confidence. On any such class that contains the Gauss and Gauss–Radau pair of Theorem 5.3 (a rich class), together with that theorem the minimax sample requirement is $N^* \asymp \sigma^2(\Delta\eta)^{-(4K-2)}$. *On smaller classes the upper bound need not be tight: if the rates are known, for example, the weights need only order* $\sigma^2(\Delta\eta)^{-2}$ *experiments. The constants* c_0 *and* C *grow quickly with* K*, through the conditioning of the moment map; the theorem gives the order in* $\Delta\eta$ *for fixed* K*, not a usable constant for large* K *(in Supplement Table S4 our double-precision Prony implementation already fails at* $K = 4$ *on the disclosure grid).*

Proof. The kernel values $k_m = \sum_i w_i r_i^m$ are the power moments of the merged measure. The closed-loop recursion $h_t = ah_{t-1} + \kappa \sum_m k_m h_{t-1-m}$ (plus κ at $t = 0$) can be solved for $k_0, \dots, k_{2K-1}$ from $h_1, \dots, h_{2K}$. Because $h_0 = \kappa$ is a known primitive, each k_m is a polynomial in $h_1, \dots, h_{m+1}$ with coefficients depending on (a, κ) (the only division is by $\kappa h_0 = \kappa^2$), so on the bounded set of responses within sup-distance 1 of the class this map is L_h-Lipschitz, with L_h depending on the class but not on $\Delta\eta$. Let E be the event $\max_{1\le t\le 2K} |\bar{h}_t - h_t| \le \sigma N^{-1/2}\sqrt{2\log(4K/\beta)}$, where $\bar{h}_t$ averages the N responses; a Gaussian tail bound and a union bound over the $2K$ estimated taps give $P(E) \ge 1 - \beta$. On E, and provided this radius is at most 1, the estimated kernel values satisfy $\max_m |\hat{k}_m - k_m| \le L_h \sigma N^{-1/2}\sqrt{2\log(4K/\beta)}$. Centring at r_c is a fixed triangular linear map, $\mu_m = \sum_{j\le m} \binom{m}{j}(-r_c)^{m-j} k_j$, and $\mu_m = (\Delta\eta)^m t_m$ with $t_m = \sum_i w_i s_i^m$. The map $F : (s, \omega) \mapsto (t_0, \dots, t_{2K-1})$ does not depend on $\Delta\eta$; its Jacobian is a confluent Vandermonde matrix with determinant $\prod_i \omega_i \prod_{i<j}(s_i - s_j)^4$, bounded away from zero on the compact class. On ordered, γ-separated nodes F is injective, because a measure with K distinct atoms is determined by its first $2K$ moments. A C^1 map that is injective on a compact set Q and has a nonsingular Jacobian there is injective on an open neighbourhood U of Q: otherwise there are $x_n \ne y_n$ with $F(x_n) = F(y_n)$ and distance to Q tending to zero, whose limits $x, y \in Q$ satisfy $F(x) = F(y)$, so $x = y$, contradicting local injectivity at x. Applying the quantitative inverse function theorem uniformly on Q, there are $\rho_0 > 0$ and $L < \infty$, uniform over the class, such that $F(U)$ contains the ball of radius ρ_0 around every point of $F(Q)$ and the single inverse $F^{-1} : F(U) \to U$ is L-Lipschitz on each such ball. An error e in μ_m is an error $e(\Delta\eta)^{-m}$ in t_m, largest at $m = 2K - 1$. Since $\Delta\eta < 1$ and the centring map is fixed, on E we get $\|\hat{t} - t\|_\infty \le C_1\sqrt{\log(4K/\beta)}\,\epsilon_N$ with C_1 equal to $\sqrt{2}L_h$ times the norm of the centring map; and because $\sigma N^{-1/2}\sqrt{2\log(4K/\beta)} \le \sqrt{2}\,\epsilon_N\sqrt{\log(4K/\beta)}$, the radius condition holds whenever $c_0 \le 1/\sqrt{2}$. When this is below ρ_0, $\hat{t}$ lies in the ball where F^{-1} is L-Lipschitz. For an input in such a ball, $x = F^{-1}(\hat{t})$ has K distinct nodes and nonzero weights, so the $K \times K$ Hankel matrix of $\hat{t}$ is nonsingular and Prony's method returns exactly x: the Prony polynomial is determined uniquely by the Hankel system, and since the K real nodes of x reproduce $\hat{t}$ it equals $\prod_i(z - \hat{s}_i)$, so its roots are real and distinct and no complex pair can appear; Prony's method

computes the algebraic inverse of the moment map up to the order of the atoms, so, listing its output in increasing order of rate (which the error bound below keeps unambiguous once $C\sqrt{\log(4K/\beta)}\,\epsilon_N < \gamma/2$), its output is $F^{-1}(\hat{t})$ and the stated bound follows with $C = LC_1$ and $c_0 = \min\{\min(\rho_0, \gamma/(2L))/C_1,\, 1/\sqrt{2}\}$, which also keeps $C\sqrt{\log(4K/\beta)}\,\epsilon_N < \gamma/2$. No separate root-perturbation bound is needed: on the ball where F^{-1} is L-Lipschitz, Prony's roots and weights are the coordinates of $F^{-1}(\hat{t})$, so the conditioning of the root step is part of L (for an explicit step-by-step analysis of Prony's method with clustered exponents see Katz et al., 2023). The matching statement requires the class to contain the Gauss and Gauss–Radau pair of Theorem 5.3 (ordered nodes with separation at least the smaller of their two minimum gaps, and weights within $[\omega_{\min}, \omega_{\max}]$ after scaling to total weight W); for any such class the orders match, with constants that depend on the class. □

In words. With K unknown groups whose memories lie within $\Delta\eta$ of each other, the number of experiments needed grows like $(\Delta\eta)^{-(4K-2)}$. For two groups, halving the spread multiplies the data needed by 64; for three groups, by 1,024. Theorem 5.3 shows that no method does better, and this theorem shows that Prony's method achieves it.

Machine check. Lean verifies this theorem for every $K \geq 1$, using only Lean's standard axioms. The formal proof follows the one above: a Lipschitz bound for the closed-loop deconvolution that does not depend on $\Delta\eta$, the $(\Delta\eta)^{-m}$ rescaling, a sub-Gaussian tail bound over the $2K$ taps, injectivity of the Jacobian of the moment map (proved by Hermite interpolation, without the determinant formula), and a uniform inverse function theorem on the compact class. The formal estimator is the representation with increasing nodes and nonzero weights that matches the $2K$ estimated moments; Lean shows that it is unique and that Prony's linear system has a unique solution whose polynomial has exactly its nodes as roots, so the roots are real. The formal statement allows sub-Gaussian noise and assumes $|r_c| + \Delta\eta \leq 1$, so that every memory rate has modulus at most one.

Table S32 confirms the order numerically. For numerical stability the estimator works in centred and rescaled coordinates (the estimator is the same; only its floating-point evaluation changes). The argument parallels the stability analysis of super-resolution for clustered complex nodes (Batenkov et al., 2021), and Katz et al. (2023) show that Prony's method attains the min-max bound for clustered exponents; here the nodes are real and are first extracted from the closed loop, and it reproduces the ESPRIT slope of -6.15 reported for $K = 2$ in the earlier version.

Real market noise. Theorems 5.3 and 5.4 assume Gaussian noise. A pre-specified test (Supplement S16) keeps the trader groups synthetic and replaces the noise with heavy-tailed one-minute Bitcoin price changes: the price-only requirement still grows with fitted slopes -6.01 for $K = 2$ and -9.81 for $K = 3$ (theory -6 and -10), while the tagged requirement stays bounded across the spacing grid.

General memory. If memory is any completely monotone kernel, a finite or infinite mixture of exponential memories, recovery becomes far harder: the minimax accuracy for the memory distribution is of exact order $1/\log(N/\sigma^2)$, so reaching accuracy w requires and suffices $N/\sigma^2 = \exp(\Theta(1/w))$ (Theorem S17.2, Supplement S17). The polynomial rates above rely on memory being a finite mixture of exponentials.

5.4 Passive data

Theorem 5.5 (Passive identification: minimum phase)**.** *In a stable economy with $0 < a < 1$, distinct $0 < r_i < 1$ and nonzero merged weights, the zeros of H are the memory rates r_i together with a structural zero at $z = 0$, and H is minimum phase. Degenerate anchor and zero-rate cases require the cancellations described in the merged-equivalence proof. With a single price innovation, the price spectrum identifies H up to scale, hence the memory rates; with κ and a known, as in Theorem 3.1, it identifies the merged measure.*

The result holds for signed weights (mean-reverting cohorts): in 1,733 stable random economies with weights of both signs, the nonstructural zeros equalled the memory rates and lay inside the unit disk. With a second independent shock in the cohorts' memories, the Jacobian of the price spectrum with respect to six model parameters has full rank, with singular values from 34.51 down to 0.0698. Passive noise helps identification rather than hindering it.

Theorem 5.6 (Passive cost)**.** *Suppose the price is observed passively as the stationary output $x_t = H_\theta(L)\xi_t$ of i.i.d. Gaussian innovations $\xi_t \sim N(0, \sigma^2)$, $\sigma^2 > 0$ known, without observation noise, for θ in a compact, uniformly Schur-stable family with rates in $[\eta_{\min}, \eta_{\max}] \subset (0, 1)$, and let $D(\theta_1 \| \theta_0)$ be the Kullback–Leibler divergence rate per observation.*

(i) For a single cohort at $\bar{\eta}$ against the symmetric pair at $\bar{\eta} \pm \delta$ with the same total weight, $D = \Theta(\delta^4)$.

(ii) For the moment-matched two-cohort economies of Theorem 5.1, $D = \Theta(\tau^2(\Delta\eta)^6)$.

(iii) For every $K \geq 2$, for the K-cohort Gauss and Gauss–Radau economies of Theorem 5.3, $D = \Theta((\Delta\eta)^{4K-2})$.

Hence any test based on T passive observations with worst-case error at most $\alpha < 1/4$ needs $T = \Omega(\delta^{-4})$, $T = \Omega(\tau^{-2}(\Delta\eta)^{-6})$ and $T = \Omega((\Delta\eta)^{-(4K-2)})$ respectively, uniformly as the separation shrinks: passive data cost at least the same polynomial order as active probing for every number of cohorts. For each fixed pair, at fixed type-I error the likelihood-ratio test's type-II error decays at exponential rate D (Stein's lemma), and both errors decay at the Chernoff-information rate, which is of the same order in the separation; so the orders are the right ones; we do not claim that this attainment is uniform as the separation shrinks.

In words. Watching the market without trading does not escape the barrier. A passive observer needs at least the same polynomial order of observations as a prober needs experiments; it pays in time rather than in trading losses. Prices recorded with extra measurement noise that is independent of the price are a garbling of the noiseless record, so by the data-processing inequality they carry less information, and the lower bound holds a fortiori.

The proof (Supplement S1) combines the Whittle form of the divergence rate with the resolvent identity $H_1 - H_0 = z^{-1}H_1H_0(G_1 - G_0)$: the two spectra differ by the leading unmatched moment, of order δ^2 for the pair and $\tau(\Delta\eta)^3$ for the moment-matched economies, and the divergence rate is the square of that difference. Numerically, the fitted slopes of the required sample length are -4.21 and -5.93, against the theoretical -4 and -6; for part (iii) the fitted slopes of the Whittle divergence rate in $\Delta\eta$ are 5.91, 9.98 and 14.05 for $K = 2, 3, 4$, against 6, 10 and 14.

Passive surveillance is possible in principle but needs at least the same polynomial order of observations as active probing needs experiments. Proposition S3.2 is the exact finite-sample version for the known pair.

6 What disclosure buys

Because active probing is costly and passive data are slow, the natural policy question is what a supervisor gains from disclosure. We compare six observation contracts at $\delta = 0.05$.

- I_0: public prices only.
- I_1: prices and aggregated executed flow.
- I_2: prices and one independent noisy report of dealer marked equity per reset ($\sigma_E = 0.5$).
- I_3: prices and exact audited equity (Proposition 4.4), also with reports rounded to 0.01.

Table 2: Disclosure contracts at $\delta = 0.05$ (pulse-design capital, par units)

Contract	Channel	Pulses	Benchmark	Saving vs. $T = 20$
I_0: prices, $T = 3$	public prices	271	4,305.25	−2,123.5%
I_0^*: prices, $T = 20$	full price path	12	193.62	0.0%
I_1: prices and signed flow	public flow	12	193.62	0.0%
I_2: noisy equity ($\sigma_E = 0.5$)	statement	12	193.62	0.0%
I_3: exact equity, 5-unit lot	audited equity	1	12.70	93.4%
I_3: exact equity, 0.01 rounding	rounded report	1	15.52	92.0%
I_4: timestamps only	no likelihood	12	193.62	0.0%
I_5: audit registry	internal states	0	0.00	100.0%

Closed-form values. Noisy equity adds information linearly in the number of resets, $D_{\mathrm{KL}} = N\Delta_E^2/(2\sigma_E^2)$, and saves only 1.5% against the $T = 3$ baseline. The equity difference is $\Delta E = \kappa^2 q \delta^2 V^2$; with reports rounded to 0.01, a guaranteed change needs $V \geq 5.59$.

- I_4: algorithm update timestamps without the update values.
- I_5: a supervisory audit registry with access to internal algorithm states.

Table 2 shows that disclosure is not one thing. Timestamps without update values carry no information about δ in this model. Noisy periodic statements barely help. Exact audited equity collapses the requirement to a single trade, limited only by lot sizes and report precision, and an audit registry removes the trading requirement altogether.

The cleanest form of useful disclosure is flow tagged by participant class.

Proposition 6.1 (Tagged flow removes the barrier)**.** *Let the price be driven by i.i.d. innovations of variance σ_ξ^2 in a positive, stable economy, with stationary autocovariances $\gamma_x(s) = \sum_\ell c_\ell \lambda_\ell^{|s|}$ ($c_\ell \geq 0$, $0 \leq \lambda_\ell < 1$ the closed-loop modes). Suppose cohort i has merged weight $w_i > 0$ and the observer receives prices and each cohort's flow $\tilde{u}_{i,t} = u_{i,t} + \varepsilon_{i,t}$, $u_{i,t} = q_i y_{i,t}$, with i.i.d. measurement noise of variance σ_f^2 independent of prices. Then $\tilde{u}_{i,t} = r_i \tilde{u}_{i,t-1} + w_i x_t + e_{i,t}$ with $e_{i,t} = \varepsilon_{i,t} - r_i \varepsilon_{i,t-1}$, and the just-identified IV estimator of $\beta_i = (r_i, w_i)$ with instruments $Z_t = (x_{t-1}, x_t)^\top$ satisfies $\sqrt{T}(\hat{\beta}_i - \beta_i) \to N(0, V_i)$,*

$$V_i = \Sigma_i^{-1} \Omega_i \Sigma_i^{-\top}, \qquad \Sigma_i = \begin{pmatrix} w_i A_0 & \gamma_x(1) \\ w_i A_1 & \gamma_x(0) \end{pmatrix}, \qquad \Omega_i = \sigma_f^2 \big[(1 + r_i^2)\Gamma_0 - r_i(\Gamma_1 + \Gamma_1^\top) \big], \quad (19)$$

where $A_m = \sum_{s \geq 0} r_i^s \gamma_x(s + m)$, $\Gamma_0 = \begin{pmatrix} \gamma_x(0) & \gamma_x(1) \\ \gamma_x(1) & \gamma_x(0) \end{pmatrix}$ and $\Gamma_1 = \mathbb{E}[Z_t Z_{t-1}^\top]$. The Jacobian determinant is

$$\det \Sigma_i = w_i \sum_{\ell, m} c_\ell c_m \frac{1 - \lambda_\ell \lambda_m}{1 - r_i \lambda_m} \; > \; 0. \quad (20)$$

V_i depends only on cohort i's own (w_i, r_i), σ_f^2 and the price autocovariances, which are bounded and nondegenerate uniformly on a uniformly stable family with merged weights

bounded below, whatever the rate spacing; so the merged weights are estimated at rate $T^{-1/2}$ with a constant that does not depend on $|r_i - r_j|$.

In words. If each group's trading is reported separately, each group becomes an ordinary regression. Its precision depends on the group's own rate and on the price dynamics, but it has no singularity as the groups' rates approach each other.

Each tagged flow follows $u_{i,t} = r_i u_{i,t-1} + w_i x_t$, a regression whose information about w_i does not involve the other cohorts' rates; the closed form (20) matches the simulated Jacobian (0.1330 against 0.1322 in a two-cohort economy with 4×10^5 periods). In simulation ($T = 5{,}000$, 60 replications) the root-mean-square error of the recovered weights is flat in $\Delta\eta$ (fitted slope 0.011, error between 5.8×10^{-4} and 6.6×10^{-4}). This is the formal content of a disclosure regime such as a consolidated audit trail: tagging flow by participant class turns a polynomial barrier into an ordinary estimation problem.

6.1 How much disclosure is worth

Proposition 6.1 says that tagging removes the barrier; the next result says by how much.

Theorem 6.2 (Value of class-tagged disclosure)**.** *Take a rich class of Theorem 5.4: rates $r_c + \Delta\eta\, s$ with s in a compact set of γ-separated ordered points that has nonempty interior and contains the Gauss and Gauss–Radau pair of Theorem 5.3, and merged weights in a nondegenerate interval $[\omega_{\min}, \omega_{\max}]$, $0 < \omega_{\min} < \omega_{\max}$. In each reset experiment the observer applies a unit impulse and records the price response with independent $N(0, \sigma^2)$ noise per tap; under tagging it also records each class's flow response $f_{i,t} = w_i \sum_{s \le t} r_i^{t-s} h_s$ with independent $N(0, \sigma_f^2)$ noise per tap. Let N_p and N_t be the numbers of experiments needed, without and with tagging, for a fixed accuracy and confidence. There are constants $0 < c \le C < \infty$, depending on (σ, σ_f) and the class but not on $\Delta\eta$, such that*

(i) to recover the weights to a fixed accuracy smaller than half the weight difference between the Gauss and Gauss–Radau economies (so that recovering the weights decides between them), $c \le N_t \le C$, so $N_p / N_t \asymp (\Delta\eta)^{-(4K-2)}$;

(ii) to recover the rates to accuracy $c_0 \Delta\eta$, for a fixed $c_0 > 0$ smaller than the radius of a ball contained in the class of rescaled nodes (so that no fixed guess meets the target), $c(\Delta\eta)^{-2} \le N_t \le C(\Delta\eta)^{-2}$, so $N_p / N_t \asymp (\Delta\eta)^{-(4K-4)}$.

Proof. $N_p \asymp \sigma^2 (\Delta\eta)^{-(4K-2)}$ by Theorems 5.3 and 5.4. *Upper bounds.* Average the N price responses to estimate h with error of order $\sigma N^{-1/2}$ per tap. Given h, class i's mean flow is a two-parameter family $F(w_i, r_i; h)$ whose Jacobian columns $\sum_s r_i^{t-s} h_s$ and $w_i \sum_s (t -$

$s) r_i^{t-s-1} h_s$ are linearly independent, with a Gram determinant bounded below uniformly on the class because $w_i \geq \omega_{\min}$, the rates lie in a compact subset of $(0,1)$ and $h_0 = \kappa > 0$; the bound does not involve the other classes or $\Delta\eta$. Least squares on class i's flows with the plug-in h therefore estimates (w_i, r_i) with error $O_P(N^{-1/2})$, uniformly on the class, which gives the upper bounds in (i) and in (ii) for accuracy $\epsilon\Delta\eta$. For (i) there is also a finite-sample version: in the first period the tagged flow of class i is $w_i\kappa$, so the mean of the N first-period records gives $P(|\hat{w}_i - w_i| \geq \epsilon) \leq \sigma_f^2/(N\kappa^2\epsilon^2)$ by Chebyshev's inequality, for every configuration (this step is machine-checked). *Lower bounds.* At a configuration in the interior of the class, perturb one class's weight by ϵ, or its rate by $\epsilon\Delta\eta$; for small ϵ both perturbed configurations remain in the class because the weight interval is nondegenerate and the rate set has nonempty interior. The mean differences of both the price and the flow records are of order ϵ, respectively $\epsilon\Delta\eta$, so the Kullback–Leibler divergence of N experiments is at most $CN\epsilon^2$, respectively $CN\epsilon^2(\Delta\eta)^2$, and Le Cam's two-point lemma gives the lower bounds. □

In words. This is the value of disclosure measured in data. Tagging replaces a requirement that explodes as the groups become similar with one that does not grow at all (for the weights) or grows only like $(\Delta\eta)^{-2}$ (for the rates).

Proposition 6.1 and Theorem 6.2 use different observation arrangements: the proposition concerns passive, stationary data with prices observed without noise, where the price is a valid instrument; the theorem concerns reset experiments with noisy prices. With noisy prices, regressing flow on the observed price is an errors-in-variables problem, which is why the theorem works from the known impulse instead.

With the same noise and lags for both observers, the Cramér–Rao requirements give fitted slopes of -6.08 for prices and -4.34 for the ratio at $K = 2$ (theory -6 and -4). Calibrated to the class memories estimated below, prices alone would need about 333 times as many days as the published class ratios to separate the top-trader and all-account memories in BTCUSDT (rates 0.9889 and 0.9986 per five minutes; about 6,000 days against 18), and about 19 times as many in ETHUSDT (rates 0.9793 and 0.9972). These rates come from day-demeaned regressions, which bias persistence down (Supplement S19), so the factors are illustrative.

6.2 When disclosure is worth acquiring

When is disclosure worth its cost? The dollar illustration of Supplement S8 prices disclosure; it does not say when a supervisor should buy it. Remark 5.2 shows that a supervisor who only needs to bound aggregate price risk does not need disclosure at all: the

summed response is identified from public prices at the ordinary rate. Attribution matters when supervision is selective: one algorithm must be stopped or sanctioned while the others keep supplying liquidity, and the law requires attributing the conduct to it at a given error level. A supervisor without an audit trail does not run trading probes; it waits for public data to accumulate, and the cost of waiting is the harm done in the meantime.

Consider M markets, a supervisory cycle of $\bar{T}$ periods, and in each market one algorithm whose conduct does harm $\ell > 0$ per period until it is attributed. With prices alone, attribution at the required error level takes $T_p(\Delta\eta) = c_p(\Delta\eta)^{-(4K-2)}$ periods; with class-tagged records it takes $T_t(\Delta\eta) = c_t(\Delta\eta)^{-2}$ periods, the orders of Theorems 5.6 and 6.2 with constants $c_p, c_t > 0$. We take the attribution times to be of exactly these orders, which is a modelling assumption: Theorem 5.6 gives only a lower bound for prices, and if price-only attribution is slower, G below is larger and disclosure is chosen at least on the band the proposition describes. $T_t \propto (\Delta\eta)^{-2}$ is the rate target; for weights at fixed accuracy T_t does not grow as $\Delta\eta$ shrinks, and the lower edge of the band disappears. Attribution that would take longer than the cycle does not happen within it. An audit trail costs F per cycle to operate and c per market to comply with, including any information leakage. The supervisor's expected loss is therefore $\mathcal{L}_{\text{price}} = M\ell \min\{\bar{T}, T_p\}$ without disclosure and $\mathcal{L}_{\text{disc}} = F + Mc + M\ell \min\{\bar{T}, T_t\}$ with it. Write $\Delta\eta_t = (c_t/\bar{T})^{1/2}$ and $\Delta\eta_p = (c_p/\bar{T})^{1/(4K-2)}$ for the spacings at which tagged and price-only attribution just fit within the cycle.

Proposition 6.3 (When an audit trail is chosen)**.** *Let $K \geq 2$ and $\Delta\eta_t < \Delta\eta_p$, and let $G(\Delta\eta) = \min\{\bar{T}, T_p(\Delta\eta)\} - \min\{\bar{T}, T_t(\Delta\eta)\}$ be the attribution delay that disclosure saves.*

(i) The supervisor strictly prefers disclosure if and only if $F/M + c < \ell\, G(\Delta\eta)$.

(ii) G vanishes for $\Delta\eta \leq \Delta\eta_t$, increases on $(\Delta\eta_t, \Delta\eta_p]$ to its maximum $G^* = \bar{T} - T_t(\Delta\eta_p) > 0$, decreases wherever it is positive beyond $\Delta\eta_p$, and is negative for $\Delta\eta > (c_p/c_t)^{1/(4K-4)}$, where it returns toward zero from below. Hence the spacings at which disclosure is chosen form an interval around $\Delta\eta_p$, possibly empty: disclosure is never chosen for algorithms so similar that even tagged records cannot attribute them within the cycle, nor for algorithms so different that prices attribute them quickly.

(iii) The interval is non-empty if and only if the audit trail covers more than $M^* = F/(\ell G^* - c)$ markets (with $M^* = \infty$ if $\ell G^* \leq c$), and it widens as M grows.

Part (i) restates the comparison of losses. Part (ii) says that disclosure is valuable in a middle band, where algorithms are close enough that prices cannot attribute them within the cycle but far enough apart that tagged records can; the polynomial gap between the two delays, $(\Delta\eta)^{-(4K-2)}$ against $(\Delta\eta)^{-2}$, is what makes the band wide. Part (iii) is an economies-of-scale result: an audit trail with a fixed operating cost pays only when it spans enough

markets. This illustrates a possible economy of scale for market-wide reporting; the cost function is stylised, not estimated from CAT, and the result neither explains CAT's legal architecture nor shows that it is optimal. A supervisor that can trade has a second option, active probing, whose cost is the frontier of Table S9; because that cost grows at the same polynomial order, it shortens price-only attribution only for well-separated algorithms and leaves the comparison for close ones unchanged. The model takes the harm ℓ as given and holds the algorithms' behaviour fixed across regimes; Section 8 returns to both.

A decision in which attribution matters. Proposition 6.3 takes the harm from slow attribution as given. The next result derives such a harm inside the model: a supervisor that must restore stability can act on the whole market, which needs only the combined response, or halt one group, which needs that group's own intensity.

Proposition 6.4 (Market-wide versus targeted intervention)**.** *In the positive, stable economy of Section 2 with known a and κ, let a supervisor require the closed-loop persistence $S = a + \kappa \sum_i q_i$ to be at most $\bar{s} \in (a, 1)$. A market-wide measure scales every group's intensity by $\theta \in (0, 1]$; a targeted measure halts one group i, removing $q_i = w_i/(1 - r_i)$.*

(i) The market-wide measure needed, $\theta^* = \min\{1, (\bar{s}-a)/(S-a)\}$, depends on the economy only through $S = a+\kappa \int (1-r)^{-1} dM(r)$, which also equals $1-\kappa/\sum_j h_j$; it is estimated at the parametric rate (Remark 5.2).

(ii) Halting group i suffices if and only if $S - \kappa q_i \le \bar{s}$. Let A and B be the two K-cohort economies of Theorem 5.3 built from the K-point Gauss and Gauss–Radau rules, and let $\omega^G_{\max} \neq \omega^R_{\max}$ be the largest normalized weights of the two rules. Then $|S^A - S^B| \le C(\Delta\eta)^{2K-1}$, while $|q^A_{\max} - q^B_{\max}| \ge W|\omega^G_{\max} - \omega^R_{\max}|/(1-r_c) - C'\Delta\eta$. Hence, for small $\Delta\eta$, there is an interval of thresholds $\bar{s}$ at which halting the largest group restores stability in one economy and not in the other, and any rule based on N reset experiments that chooses correctly with probability at least $3/4$ in both economies needs, with $z_{0.75} = \Phi^{-1}(0.75)$,

$$N \;\ge\; \Big(\frac{2 z_{0.75}\,\sigma}{\|h^A - h^B\|_2}\Big)^2 \;\ge\; c\,\sigma^2 (\Delta\eta)^{-(4K-2)}.$$

(iii) With class-tagged flow as in Theorem 6.2, each q_i is estimated to any fixed accuracy from a number of experiments that does not depend on $\Delta\eta$, so for a fixed decision margin $|S - \kappa q_i - \bar{s}| > 0$ the targeted decision needs a bounded number of experiments.

Proof. (i) Scaling all intensities by θ gives persistence $a + \theta(S - a)$, which is at most $\bar{s}$ exactly when $\theta \le (\bar{s} - a)/(S - a)$; $q_i = w_i/(1 - r_i)$ gives the integral form, and $H(1) =$

$\kappa/(1-S) = \sum_j h_j$ gives the second. (ii) Write $f(r) = (1-r)^{-1}$ and expand around r_c: $f(r_c + \Delta\eta s) = \sum_m (1-r_c)^{-(m+1)}(\Delta\eta s)^m$. The two economies have the same total weight W and the same moments of order $0, \dots, 2K-2$ in $s \in [-1,1]$, so $|S^A - S^B| \le 2\kappa W \sum_{m \ge 2K-1} (\Delta\eta)^m (1-r_c)^{-(m+1)} \le C(\Delta\eta)^{2K-1}$ for $\Delta\eta \le (1-r_c)/2$. Each $q_i = W\omega_i/(1 - r_c - \Delta\eta s_i)$ is within $W\omega_i \Delta\eta/((1-r_c)(1-r_c-\Delta\eta))$ of $W\omega_i/(1-r_c)$, which gives the bound on the largest intensities. Let A be the economy with the larger $q_{\max}$. Then $(S^B - \kappa q^B_{\max}) - (S^A - \kappa q^A_{\max}) > 0$ for small $\Delta\eta$, and any $\bar{s}$ strictly between the two post-halt persistences, and below both S^A and S^B, separates the correct decisions. A rule that is correct with probability at least 3/4 in both economies is a test between them with both errors at most 1/4. For Gaussian observations with a common covariance the best such test has error $\Phi(-\sqrt{N}\|h^A - h^B\|_2/(2\sigma))$, which gives the first inequality, and $\|h^A - h^B\|_2 \asymp (\Delta\eta)^{2K-1}$ (Theorem 5.3) gives the second. (iii) As in the proof of Theorem 6.2, tagged flow gives each (w_i, r_i) with error of order $N^{-1/2}$ uniformly in the spacing, and $\eta_i = 1 - r_i$ is bounded away from zero on the class. □

In words. A supervisor that only needs to cool the whole market can act on public data: the size of a market-wide measure depends on the combined response, which is learned quickly. A supervisor that wants to stop one group and leave the others trading needs that group's own trading intensity, and there are two markets that look almost the same in prices, one in which halting the largest group is enough and one in which it is not. Telling them apart takes the full $(\Delta\eta)^{-(4K-2)}$ cost; class-tagged records reduce it to a bounded number of experiments. Table S33 gives numbers. The largest normalized weights of the two rules differ for every $K \le 8$ that we checked (for example 0.500 against 0.750 for $K = 2$ and 0.444 against 0.512 for $K = 3$); we do not have a proof that they differ for every K. This proposition is new to this revision and has not been independently reviewed.

What the data can and cannot test. The theorems concern experiments that no outside researcher can run: reset initial states, known price persistence and impact, and either controlled probes or stationary data from a known innovation law. Public data meet none of these exactly. Order flow is endogenous, the fundamental value is unobserved and replaced by returns, observations overlap in time, parameters drift, and the class reports are positions aggregated over many accounts rather than tagged flow. The tests below therefore check the theory's observable prediction, that class records restrict persistence which prices leave unrestricted, under a stated mapping from class positions to cohort states. They do not estimate the theorems' rates, and a pass does not show that class persistence is learning memory.

6.3 Evidence from trader-class reports

Two public data sets report positions by trader class: the CFTC's weekly Commitments of Traders, mandated by law, in eight regulated futures markets, and Binance's five-minute top-trader and all-account ratios, published voluntarily, in 30 perpetual markets. Supplements S18 and S19 report the pre-specified tests in full. In both, class positions pin each class's fitted persistence to a narrow interval while prices leave it essentially unrestricted: the primary CFTC hypothesis holds in all eight markets, with price-only intervals 9 to 71 times wider, and on 30 fresh Binance markets the horizon-matched comparison gives a median width ratio of 7.3. Several pre-specified tests failed and are reported as they came out (Table 1): Binance class data did not improve out-of-sample return forecasts, the CFTC reports read as flows pinned down their own persistence in only 3 of 8 markets (6 were required), and a test designed to separate learning memory from position turnover found a distinct faster root in only 2 of 8. A fitted persistence of aggregated positions also reflects how slowly positions turn over, so these data restrict how persistent class positions are, not learning memory itself.

7 Implications for supervision and market design

The results separate two supervisory tasks that are often run together. Monitoring aggregate fragility, for example to set market-wide limits, needs only the combined price response, which public data identify at the ordinary rate however many groups trade (Remark 5.2, Proposition 6.4(i)). Attributing feedback to particular groups, to apply targeted limits or enforcement, faces the $(\Delta\eta)^{-(4K-2)}$ barrier whenever groups react over similar horizons (Proposition 6.4(ii)). Class-tagged records remove that barrier group by group (Proposition 6.1), and in the model persistent pseudonymous class labels deliver the full statistical gain without personal data.

What a supervisor can learn from public data. Prices and anonymous flow reveal the merged measure M and the closed-loop impulse response, hence the transfer function, and nothing finer. A supervisor who wants to know *which* algorithms drive a feedback loop cannot get there from public data at reasonable cost when their memories are close. This is not a failure of econometric technique: Theorem 5.1 holds for every estimator, biased or not.

What probing costs. At realistic observation noise, an exchange or a large desk could separate two well-separated memory rates for about 10^5 USDT of capital. Probing is there-

fore feasible for coarse questions and infeasible for fine ones. Because capital grows as δ^{-4} and experiments as $(\Delta\eta)^{-(4K-2)}$, halving the separation multiplies the cost by 16 for a known pair and by 2^{4K-2} for K unknown cohorts.

What disclosure buys. Disclosure regimes differ sharply in value (Table 2). Timestamps without update values are worthless for this purpose. Noisy periodic statements are nearly worthless. Audited equity and, above all, flow tagged by participant class remove the barrier entirely (Proposition 6.1). In our attribution experiment, correctly tagged flow is far more informative than the periodic statements considered here. Calibrated to the fitted day-demeaned persistences of Binance's published class ratios (an illustrative calibration), separating the top-trader group from all accounts in BTCUSDT would take prices alone about 333 times as long (Theorem 6.2). The class reports that the CFTC already requires have the same property in regulated markets (Table S24), so the argument concerns the granularity and frequency of existing disclosure, not a new kind of reporting.

When audit trails already exist. The US Consolidated Audit Trail (SEC Rule 613) is designed to capture specified order and reportable-event information for NMS stocks and listed options across exchanges and broker-dealers, from origination to execution or cancellation, linked to a customer identifier and to the broker-dealer that handled the order; its actual scope is governed by Rule 613, the CAT NMS Plan and exemptions. Since February 2025, under SEC exemptive relief, it no longer requires the names, addresses and years of birth of natural-person customers, who remain linked through an identifier derived from a transformed tax identifier (SEC, 2012, 2016, 2024, 2025). Our results bear on such regimes in three conditional ways. First, for the attribution task modelled here, records that report each class's flow separately remove a barrier that, for an observer who learns by probing, costs round-trip trading loss of order δ^{-4}, and they save samples by a factor of order $(\Delta\eta)^{-(4K-2)}$ at fixed accuracy or $(\Delta\eta)^{-(4K-4)}$ for rates resolved to a fraction of their spacing (Proposition 6.1, Theorem 6.2). This is a measurement value, not a cost–benefit assessment: CAT's budget pays for enforcement, market reconstruction and surveillance, none of which the model measures, and Supplement S8 sets the hypothetical probing loss beside it only as a scale illustration. For estimating class responses in our model a persistent pseudonymous label suffices; this does not settle what enforcement or data-minimization design requires. Second, an audit trail records accounts, not decision rules. One account can run several algorithms and one algorithm can trade through several accounts, so account-level records need not provide the class-homogeneous flow that Proposition 6.1 assumes; an observer who wants rule-level memories faces our problem within each account's flow, with fewer cohorts but the

same exponent. MiFID II's order-record and algorithmic-governance rules (RTS 24 for venue order records, European Commission, 2017; RTS 6 for firms' algorithmic systems, European Commission, 2017b) are the closer analogue, although the firm-assigned algorithm identifier tracks a distinct strategy within a firm, not its continuously updated parameters. Third, the audit trail does not reach the markets we study. Crypto perpetual futures are outside it, futures regulated by the CFTC have separate reporting, and its data are not public: regulators and authorized self-regulatory organizations have controlled access, while outside desks and researchers generally do not. For these observers the price-only and coarse-disclosure contracts of this paper are the relevant ones.

What to monitor instead. Because transient loss can be arbitrarily large at fixed eigenvalues and volatility (Theorem S12.1), supervision that wants to control risk rather than identify participants should monitor quantities that bound transient loss directly: the summed absolute impulse response and confidence sets for the transfer function, both of which are identified from public data. These are cheap where identification is expensive (Remark 5.2); the companion paper shows that the summed response can be bounded at the parametric rate from controlled experiments, and only under a declared bound on memory length from a finite window (Niroula et al., 2026).

How to measure feedback. Empirical feedback indices should (i) compare persistence with the mechanical value of the filter, (ii) align flow with the bar in which it is observed, (iii) control for order-splitting persistence or use instruments with a defended exclusion, and (iv) report placebo pipelines. Each of these changes the sign or the order of magnitude of published-style estimates in our data.

8 Conclusion and an open problem

In a linear market with exponential-memory cohorts and known impact parameters, prices reveal a merged summary of algorithmic feedback. Recovering the algorithms behind it costs executed quantity of order δ^{-2} and trading capital of order δ^{-4} for a known pair, with a numerically evaluated constant of about 61 par units at the calibration, and a number of experiments of exact order $(\Delta\eta)^{-(4K-2)}$ for K unknown cohorts, and at least that order of passive observations; in simulations of stylized nonlinear rules and of online learners trained in the closed loop the same costs hold when the competing populations' feedback weights match, populations whose learned weights differ are told apart by those weights instead, and hard thresholds hide cohorts from small probes altogether. Without the exponential

form, accuracy improves only as $1/\log N$, with matching lower and upper bounds. Class-tagged disclosure removes the barrier under the stated contract, and its value can be stated in samples: a factor of order $(\Delta\eta)^{-(4K-4)}$ for resolving rates (and $(\Delta\eta)^{-(4K-2)}$ at fixed accuracy), about 333 for the trader classes Binance already publishes in BTCUSDT. The CFTC's class reports restrict a fitted position persistence that prices leave unrestricted in all eight regulated markets we examined; this persistence is not shown to be learning memory, a pre-specified test of what prices do reveal failed, and class-tagged memory states did not improve 30-minute post-shock return forecasts ($\Delta R^2 = -1.09$ points, inconclusive). Read as weekly flows, the same CFTC reports identify how each class's flow responds to prices but failed a pre-specified test of how precisely they pin down its persistence, and a pre-specified test for a second, faster root in positions, which would have been the learning memory, failed as well. With real Bitcoin price noise in place of Gaussian noise, the predicted rates held in a pre-specified test, at the sample sizes where the real noise could be used directly. At realistic noise the capital barrier for a moderate separation is about 10^5 USDT. Measured feedback in crypto markets is real but small, rises with depth, does not move when fees change trading volume severalfold, and is easily confused with filtering and timing artefacts. Finally, eigenvalues and volatility cannot certify safety, because transient loss can be arbitrarily large at fixed spectrum and variance.

Two open questions remain. The first is economic: Proposition 6.4 gives one stylised decision in which knowing the individual groups changes the right action, but turning the information costs here into a value of disclosure needs a supervisory loss function estimated from real interventions. The second is for supervision: *what can be measured and controlled without identifying the algorithms?* The impulse response and the transfer function, and finite-sample confidence sets for them, are identified from public data even when cohorts are not. The structural matrix J is not: economies with the same merged measure can have different state dimensions, so J is identified only up to the equivalence class of minimal realizations of the same transfer function. A well-posed next question is: for a stated class of stable closed-loop markets, given an estimated transfer function and a finite-sample confidence set, which intervention policy of least cost guarantees a stated bound on future price displacement without recovering individual cohorts? The companion paper uses exactly this interface to build an identification-free pre-trade control.

Declarations

Funding. This research received no external funding.

Competing interests. The authors declare no competing interests.

Author contributions. Pratik Niroula: conceptualization, methodology, software, formal analysis, writing (original draft). Prashant Niroula: formal analysis (verification of proofs), validation, writing (review and editing). Sujan Khatiwada: formal analysis (verification of proofs), validation, writing (review and editing). All authors read and approved the final manuscript.

Data availability. All data are public. Binance market data come from the Binance public data archive (`https://data.binance.vision`): spot and USD-margined perpetual futures candles at one-second and one-minute frequency, futures order-book depth, and the published trader-class long/short ratios (top traders by position and by account, and all accounts); every file is verified against Binance's published checksums. The CFTC Commitments of Traders reports (Traders in Financial Futures and Disaggregated, futures only, through 13 October 2026) come from the CFTC public reporting API. Weekly prices for the CFTC markets (stock index, currencies, Treasury yield and CME futures) come from Yahoo Finance. Processed data and all result files are in the public repository below.

Code availability. All code, pre-specified test specifications, result files and test-unlock counters are public at `https://github.com/Pratikn03/Fianince`. The repository's README lists the scripts in run order, and Supplement S23 maps every number in the paper to the file that produces it.

Ethics approval. Not applicable. The study uses only public market data and involves no human participants or personal data.

Use of AI tools. The research ideas are the authors' own. The authors used AI assistants in carrying them out: Claude (Anthropic) was used to analyse data, read, write and debug analysis code, check derivations and numerical results, and draft and edit text. ChatGPT (OpenAI) and Gemini (Google) were used for planning and to critique drafts; these AI critiques are internal checks, not independent peer review. AI tools are not authors. The authors ran and reviewed all code and results, reviewed all text, and take full responsibility for the content.

References

Alfonsi, A., A. Schied, and A. Slynko (2012). Order book resilience, price manipulation, and the positive portfolio problem. *SIAM Journal on Financial Mathematics* 3. doi:10.1137/110822098.

Alfonsi, A., A. Schied, and F. Klöck (2016). Multivariate transient price impact and matrix-valued positive definite functions. *Mathematics of Operations Research* 41, 914–934.

Almgren, R., and N. Chriss (2001). Optimal execution of portfolio transactions. *Journal of Risk* 3, 5–39.

Amilon, H. (2008). Estimation of an adaptive stock market model with heterogeneous agents. *Journal of Empirical Finance* 15, 342–362.

Andrews, D. W. K., and X. Cheng (2012). Estimation and inference with weak, semi-strong, and strong identification. *Econometrica* 80, 2153–2211.

Banerjee, S., and M. Szydlowski (2026). Trading against algorithms: Price dynamics and risk-sharing in a market with Q-learners. Working paper.

Barberis, N., R. Greenwood, L. Jin, and A. Shleifer (2015). X-CAPM: An extrapolative capital asset pricing model. *Journal of Financial Economics* 115, 1–24.

Barberis, N., R. Greenwood, L. Jin, and A. Shleifer (2018). Extrapolation and bubbles. *Journal of Financial Economics* 129, 203–227.

Barth, D., and R. J. Kahn (2021). Hedge funds and the Treasury cash-futures disconnect. Office of Financial Research Working Paper 21-01.

Batenkov, D., G. Goldman, and Y. Yomdin (2021). Super-resolution of near-colliding point sources. *Information and Inference* 10, 515–572.

Bernstein, S. (1914). Sur la meilleure approximation de $|x|$ par des polynômes de degrés donnés. *Acta Mathematica* 37, 1–57.

Bessembinder, H., W. Maxwell, and K. Venkataraman (2006). Market transparency, liquidity externalities, and institutional trading costs in corporate bonds. *Journal of Financial Economics* 82, 251–288.

Biais, B., T. Foucault, and S. Moinas (2015). Equilibrium fast trading. *Journal of Financial Economics* 116, 292–313.

Boehmer, E., G. Saar, and L. Yu (2005). Lifting the veil: An analysis of pre-trade transparency at the NYSE. *Journal of Finance* 60, 783–815.

Bouchaud, J.-P., Y. Gefen, M. Potters, and M. Wyart (2004). Fluctuations and response in financial markets: The subtle nature of "random" price changes. *Quantitative Finance* 4, 176–190.

Bouchaud, J.-P., J. Bonart, J. Donier, and M. Gould (2018). *Trades, Quotes and Prices: Financial Markets Under the Microscope*. Cambridge University Press.

Brock, W. A., and C. H. Hommes (1998). Heterogeneous beliefs and routes to chaos in a simple asset pricing model. *Journal of Economic Dynamics and Control* 22, 1235–1274.

Brogaard, J., T. Hendershott, and R. Riordan (2014). High-frequency trading and price discovery. *Review of Financial Studies* 27, 2267–2306.

Brunnermeier, M. K., and L. H. Pedersen (2009). Market liquidity and funding liquidity. *Review of*

Financial Studies 22, 2201–2238.

Castelli, V., and T. M. Cover (1996). The relative value of labeled and unlabeled samples in pattern recognition with an unknown mixing parameter. *IEEE Transactions on Information Theory* 42, 2102–2117.

Consolidated Audit Trail, LLC (2026). Consolidated Audit Trail, LLC 2026 Financial and Operating Budget, 31 March 2026 (total expenses $147,044,869). https://www.catnmsplan.com/cat-financial-and-operating-budget.

Cespa, G., and X. Vives (2015). The beauty contest and short-term trading. *Journal of Finance* 70, 2099–2154.

Cochrane, D., and G. H. Orcutt (1949). Application of least squares regression to relationships containing auto-correlated error terms. *Journal of the American Statistical Association* 44, 32–61.

Colliard, J.-E., T. Foucault, and S. Lovo (2022). Algorithmic pricing and liquidity in securities markets. CEPR Discussion Paper 17606.

Crouzet, N., I. Dew-Becker, and C. G. Nathanson (2020). On the effects of restricting short-term investment. *Review of Financial Studies* 33, 1–43.

Dou, W. W., I. Goldstein, and Y. Ji (2025). AI-powered trading, algorithmic collusion, and price efficiency. NBER Working Paper 34054.

Duarte, J., and L. Young (2009). Why is PIN priced? *Journal of Financial Economics* 91, 119–138.

Easley, D., N. M. Kiefer, M. O'Hara, and J. B. Paperman (1996). Liquidity, information, and infrequently traded stocks. *Journal of Finance* 51, 1405–1436.

European Securities and Markets Authority (2016). Guidelines on transaction reporting, order record keeping and clock synchronisation under MiFID II. ESMA/2016/1452.

European Commission (2017). Commission Delegated Regulation (EU) 2017/580 of 24 June 2016 supplementing Regulation (EU) No 600/2014 with regard to regulatory technical standards for the maintenance of relevant data relating to orders in financial instruments (RTS 24), Article 2(1)(b)–(c). *Official Journal of the European Union* L 87, 31.3.2017.

European Commission (2017). Commission Delegated Regulation (EU) 2017/589 of 19 July 2016 supplementing Directive 2014/65/EU with regard to regulatory technical standards specifying the organisational requirements of investment firms engaged in algorithmic trading (RTS 6). *Official Journal of the European Union* L 87, 31.3.2017.

Fan, J. (1991). On the optimal rates of convergence for nonparametric deconvolution problems. *Annals of Statistics* 19, 1257–1272.

Forssell, U., and L. Ljung (1999). Closed-loop identification revisited. *Automatica* 35, 1215–1241.

Franke, R., and F. Westerhoff (2012). Structural stochastic volatility in asset pricing dynamics: Estimation and model contest. *Journal of Economic Dynamics and Control* 36, 1193–1211.

Gatheral, J. (2010). No-dynamic-arbitrage and market impact. *Quantitative Finance* 10, 749–759.

Glosten, L. R., and P. R. Milgrom (1985). Bid, ask and transaction prices in a specialist market with heterogeneously informed traders. *Journal of Financial Economics* 14, 71–100.

Hasbrouck, J. (1991). Measuring the information content of stock trades. *Journal of Finance* 46, 179–207.

Heinrich, P., and J. Kahn (2018). Strong identifiability and optimal minimax rates for finite mixture estimation. *Annals of Statistics* 46, 2844–2870.

Hendershott, T., C. M. Jones, and A. J. Menkveld (2011). Does algorithmic trading improve liquidity? *Journal of Finance* 66, 1–33.

Hua, Y., and T. K. Sarkar (1990). Matrix pencil method for estimating parameters of exponentially damped/undamped sinusoids in noise. *IEEE Transactions on Acoustics, Speech, and Signal Processing* 38, 814–824.

Katz, R., N. Diab, and D. Batenkov (2023). On the accuracy of Prony's method for recovery of exponential sums with closely spaced exponents. arXiv preprint arXiv:2302.05883.

Kirilenko, A., A. S. Kyle, M. Samadi, and T. Tuzun (2017). The Flash Crash: High-frequency trading in an electronic market. *Journal of Finance* 72, 967–998.

Kong, W., and G. Valiant (2017). Spectrum estimation from samples. *Annals of Statistics* 45, 2218–2247.

Kreiss, H.-O. (1962). Über die Stabilitätsdefinition für Differenzengleichungen die partielle Differentialgleichungen approximieren. *BIT Numerical Mathematics* 2, 153–181.

Kukacka, J., and J. Barunik (2017). Estimation of financial agent-based models with simulated maximum likelihood. *Journal of Economic Dynamics and Control* 85, 21–45.

Kyle, A. S. (1985). Continuous auctions and insider trading. *Econometrica* 53, 1315–1335.

Le Cam, L. (1986). *Asymptotic Methods in Statistical Decision Theory*. Springer.

Leach, J. C., and A. N. Madhavan (1993). Price experimentation and security market structure. *Review of Financial Studies* 6, 375–404.

Liu, P., and H. Ammari (2022). Super-resolution of positive near-colliding point sources. arXiv preprint arXiv:2212.00536.

Ljung, L. (1999). *System Identification: Theory for the User*, 2nd ed. Prentice Hall, Upper Saddle River, NJ.

Makarov, I., and A. Schoar (2020). Trading and arbitrage in cryptocurrency markets. *Journal of Financial Economics* 135, 293–319.

Manole, T., and N. Ho (2022). Refined convergence rates for maximum likelihood estimation under finite mixture models. *Proceedings of the 39th International Conference on Machine Learning*, PMLR 162.

Manski, C. F. (2003). *Partial Identification of Probability Distributions*. Springer, New York.

Moallemi, C. C., B. Park, and B. Van Roy (2008). Strategic execution in the presence of an uninformed arbitrageur. arXiv preprint arXiv:0801.3001.

Moitra, A. (2015). Super-resolution, extremal functions and the condition number of Vandermonde matrices. In *Proceedings of the 47th Annual ACM Symposium on Theory of Computing (STOC)*, 821–830.

Moskowitz, T. J., Y. H. Ooi, and L. H. Pedersen (2012). Time series momentum. *Journal of Financial Economics* 104, 228–250.

Niroula, P., S. Alkhushayni, Y. Chen, M. Bhavsar, and P. Niroula (2026). From human markets to machine markets: Transient amplification, an exact depth condition, and a maximal pre-trade guard. Working paper, Minnesota State University, Mankato.

Obizhaeva, A. A., and J. Wang (2013). Optimal trading strategy and supply/demand dynamics. *Journal of Financial Markets* 16, 1–32.

Pagano, M., and A. Röell (1996). Transparency and liquidity: A comparison of auction and dealer markets with informed trading. *Journal of Finance* 51, 579–611.

Park, B., and B. Van Roy (2015). Adaptive execution: Exploration and learning of price impact. *Operations Research*, doi:10.1287/opre.2015.1415.

Rebafka, T., and F. Roueff (2015). Nonparametric estimation of the mixing density using polynomials. arXiv preprint arXiv:1002.4516.

Rivlin, T. J. (1981). *An Introduction to the Approximation of Functions*. Dover.

Roll, R. (1984). A simple implicit measure of the effective bid-ask spread in an efficient market. *Journal of Finance* 39, 1127–1139.

Roy, R., and T. Kailath (1989). ESPRIT—estimation of signal parameters via rotational invariance techniques. *IEEE Transactions on Acoustics, Speech, and Signal Processing* 37, 984–995.

U.S. Securities and Exchange Commission (2010). Risk management controls for brokers or dealers with market access. Release No. 34-63241.

U.S. Securities and Exchange Commission (2012). Consolidated Audit Trail; Final Rule (Release No. 34-67457, July 18, 2012, 77 FR 45722 at 45790–45795, pp. 220–238; proposing release No. 34-62174, May 26, 2010, 75 FR 32556 at 32598, 32600, pp. 147, 153).

U.S. Securities and Exchange Commission (2016). Joint Industry Plan; Order Approving the National Market System Plan Governing the Consolidated Audit Trail. Release No. 34-79318, November 15, 2016, 81 FR 84696 at 84801, p. 363.

U.S. Securities and Exchange Commission (2024). Notice of Filing of Proposed Rule Change Relating to the 2025 CAT Budget. Release No. 34-101901, December 12, 2024, p. 4 (revised May 2025).

U.S. Securities and Exchange Commission (2025). Order granting exemptive relief from certain requirements of the National Market System Plan governing the Consolidated Audit Trail. Release No. 34-102386, February 10, 2025.

Shahram, M., and P. Milanfar (2005). On the resolvability of sinusoids with nearby frequencies in the presence of noise. *IEEE Transactions on Signal Processing* 53, 2579–2588.

Shleifer, A., and R. W. Vishny (1992). Liquidation values and debt capacity: A market equilibrium approach. *Journal of Finance* 47, 1343–1366.

Sornette, D., S. Lera, J. Lin, and K. Wu (2023). Non-normal interactions create socio-economic bubbles. *Communications Physics* 6, 261.

Staiger, D., and J. H. Stock (1997). Instrumental variables regression with weak instruments. *Econometrica* 65, 557–586.

Trebbi, F., M. B. Zhang, and M. Simkovic (2024). The cost of regulatory compliance in the United States. Working paper.

Trefethen, L. N., and M. Embree (2005). *Spectra and Pseudospectra: The Behavior of Nonnormal Matrices and Operators*. Princeton University Press.

Tremacoldi-Rossi, P. (2025). Manipulating algorithmic markets. Working paper.

Tsybakov, A. B. (2009). *Introduction to Nonparametric Estimation*. Springer.

van Waarde, H. J., J. Eising, H. L. Trentelman, and M. K. Camlibel (2020). Data informativity: A new perspective on data-driven analysis and control. *IEEE Transactions on Automatic Control* 65, 4753–4768.

Varga, R. S., and A. J. Carpenter (1985). On the Bernstein conjecture in approximation theory. *Constructive Approximation* 1, 333–348.

Wagenmaker, A., and K. Jamieson (2020). Active learning for identification of linear dynamical systems. In *Proceedings of the 33rd Conference on Learning Theory*, PMLR 125, 3487–3582.

Wagenmaker, A. J., M. Simchowitz, and K. Jamieson (2021). Task-optimal exploration in linear dynamical systems. In *Proceedings of the 38th International Conference on Machine Learning*, PMLR 139, 10641–10652.

Wellman, M. P. (2006). Methods for empirical game-theoretic analysis. In *Proceedings of the 21st National Conference on Artificial Intelligence (AAAI-06)*, 1552–1555.

Whittle, P. (1953). Estimation and information in stationary time series. *Arkiv för Matematik* 2, 423–434.

Wu, Y., and P. Yang (2020). Optimal estimation of Gaussian mixtures via denoised method of moments. *Annals of Statistics* 48, 1981–2007.